

PLC Interface.

机床产品/Machine Tool Products

Exported on 02/10/2021

Table of Contents

1	CNC and PLC Interface	3
1.1	C Bit Interface (From PLC To CNC)	3
1.2	S Bit Interface (From CNC To PLC)	42
1.3	PC Keyboard C/S Mapping	60
1.3.1	Basic Key Area	60
1.3.2	Extension Key Area.....	61
1.4	R Register Interface	62
1.4.1	Resources Distribution Table	62
1.4.2	Interface Instructions.....	64

1 CNC and PLC Interface

1.1 C Bit Interface (From PLC To CNC)

No	Title	Description
C000	Cycle Start	Under auto mode, when this bit ON CNC will start to execute the NC program.
C001	Feed Hold	During machining, when this flag ON, CNC will pause the execution of the NC program.
C002		
C003		
C004		
C005		
C006®	X Axis JOG+	Under JOG MODE, when this flag ON, machine will move in X+ direction.
C007®	X Axis JOG -	(As above)
C008®	Y Axis JOG +	(As above)R
C009®	Y Axis JOG -	(As above)
C010®	Z Axis JOG+	(As above)
C011®	Z Axis JOG -	(As above)

No	Title	Description
C012®	C(6 th) Axis JOG +	(As above)
C013®	C(6 th) Axis JOG -	(As above)
C014		
C015	M30 support program rewind	This Bit is valid only when Pr732 is set to 1. When this Bit is ON, main program does not execute Reset after M30 is executed. Instead, like M99 returns to the program initial position, then continues executing machining. <u>Notifications (While Pr732 set to 1) :</u> 1. When M30 is executed, M code content description is updated in R1/R616/R620/R624. a. If C15 ON and Pr3804 part count M code number is 30, only when M30 is executed, each path will update the M code content to 30 in R1/R616/R620/R624 respectively. b. If C15 OFF, regardless of the Pr3804 part count M code number, when M30 is executed, each path will update the M code content to 30 in R1/R616/R620/R624 respectively. 2. Effective vision : 10.116.54, 10.117.54
C016	X Axis MPG Selection	Select MPG to control X axis, when this FLAG is ON, controller will input as MPG and drive the X-axis to move relative displacement.
C017	Y Axis MPG Selection	(As above)
C018	Z Axis MPG Selection	(As above)

No	Title	Description
C019	C(6 th) Axis MPG Selectio n	(As above)
C020	MPG Simulati on	1. In auto mode (Auto or MDI), when this Bit ON and cycle start, the machine axial moving speed determined by the MPG rotation speed and the current G01 override. The faster MPG is rotated, the faster the machine moves; when MPG stops, the machine also stopped. It is suitable for trial machining. 2. As mentioned above, "determined by current G01 override" means that even when the MPG is rotated under rapid traverse G00 block, and the rapid traverse feedrate (R18) is zero, machine will still move. 3. When using MPG simulation at G00 or G53 block, the maximum feedrate will be limited to the last G00 moving speed multiplied by 37.5% of the G01 override. 4. When using MPG simulation, the number of lines/single blocks to be reversed is limited. As for the number of lines/single blocks can be reversed, it depends on how the NC code is written.
C021	Tool Withdra w	When this function is triggered, the system will automatically pause (Feedhold). After the machine stops, the tool will retract according to the tool retraction position set by G10.6 and feedrate override.
C022		
C023	Rapid Travers e	Press this button under CON JOG mode, the machine movement speed changes from the JOG SPEED to RAPID SPEED(G00) , and enable the panel RAPID OVERRIDE.
C024		
C025	Set X Axis Machine Position	To set the axial machine coordinates, the set value is determined by R38.
C026	Set Y Axis Machine Position	(As above)

No	Title	Description
C027	Set Z Axis Machine Position	(As above)
C028	Set C(6 th) Axis Machine Position	(As above)
C029		
C030		
C031	X Axis MANUAL CONTROL	The axial trips from control mode and enters feedback monitoring mode. At this moment, controller can no longer send commands to the axis. If the axis moves due to external force, the HMI displayed coordinates will also be updated. Serial axis, if you need to SERVO_OFF certain axis, you can use this Cbit to SERVO OFF the serial axis. <u>Notification:</u> When the axis is switched back to the control mode from the feedback monitoring mode, if machine is still moving, will cause zero speed check fails (Pr901~), system will issue MOT-020/MOT-021 alarm, and at this moment axial needs to re-do homing.
C032	Y Axis MANUAL CONTROL	(As above)
C033	Z Axis MANUAL CONTROL	(As above)

No	Title	Description
C034	C(6 th) Axis MANUA L CONTR OL	(As above)
C035	Absolut e Position Sync.	If any manual displacement (by Jog, MPG, PLC axis...) requires to be preserved during Feedhold state, upper edge trigger this C-BIT before the program started. In this case, the amount of displacement of all axis will be synchronized to MPG offset, the program coordinates will resume to state before feedhold. And once program starts, the machining path will always retain this amount of displacement. <u>Restrictions:</u> This function is invalid when BIT triggered during machining, tool retract function executing or O888801 executing. <u>Compatibility:</u> Version 10.114.50G ~ 10.116.12 or later Between Version 10.114.50G~10.116.12, when using upper edge triggers this BIT, the main system path will perform a coordinate synchronization. This BIT has no effect in earlier versions.
C036	Emerge ncy Stop	When the Emergency Stop button is pressed, LADDER should triggers this FLAG and CNC stops all the movement of the machine and enters NOT READY state. Until the E-stop button is released, this FLAG is disabled and CNC generate the REST command and return to the READY state.
C037	External Reset	- Send the RESET command to the CNC from external device. - In 10.118.40 and later version, when Pr741 is set to 1, the reset range is not including the axis group of independent operation.
C038	M.S.T Finish	When LADEER executes M, S, T CODE, CNC will wait for this FLAG ON to start the execution of the next BLOCK. Therefore, after LADDER executed M, S, T CODE, this FLAG needs to be ON.
C039	Part count M-code stop request	When this BIT is ON, the part count M-code of the main program will be paused (B-STOP). If the part count M-code of the main program is M99, CNC will only be paused (M-99) when the M99 with the smallest path number is read. Relative reference: Parameter3804, S35

No	Title	Description
C040	Single Block	When the machining program is executed under this FLAG ON, a block will be executed and then enter the single block stop. It is necessary to Cycle Start again before executing the next single block (Block). (Please turn ON C40 before machining. If you turn ON C40 during machining, it may requires several blocks to decelerate to zero before block stop. If it is required to check the program during machining, it is recommended to stop with the MPG simulation C20 or M00.
C041	Optional Skip	When the machining program is executed while this FLAG ON, if the jump character '/' is written in the program, the line will be skipped and the next block will be executed directly. (This function cannot support partial syntax. When an unsupported syntax is encountered, a COM alarm will be issued.)
C042	Dry Run	When this FLAG is ON, the execution of G CODE is performed at an dry run speed. G01 dry run speed is up to 500% of the feedrate command in the program. The dry run speed is limited by Pr405 and Pr621~. Under this mode, G01 override is valid. G00 dry run speed is 100% of the original G00 speed.
C043	Machine Lock	When this FLAG is ON, and program is executed, the program will run but the machine will not move. It is usually used for program check. Note: Version 10.114.45, 10.115.46B, 10.115.48 and after C43 function is cancelled. This function enables (C43 ON) first, and sends a command to the axis, i.e. : G91G00X100., then axis does not move, and the remaining distance and program coordinates are displayed normally, but the machine coordinates are not changed. The system will synchronize the machine coordinates to the program coordinates until this function is turned off. If the function is enabled (C43 ON) during axial movement, the axis will not move and the remaining distance and program coordinates will be displayed normally, but the machine coordinates will stop when the function is activated. The system will synchronize the machine coordinates to the program coordinates until this function is turned off.
C044	Optional Stop	When this FLAG is ON and M01 is executed, controller will enter the FEEDHOLD state. When this FLAG is OFF and M01 is executed, controller will ignore M01 and continue to execute the next BLOCK.

No	Title	Description
C045	Z Axis Lock	When this FLAG is ON, and program is executed, the program will run but the machine will not move. It is usually used for program check. This function enables (C45 ON) first, and sends a command to the Z axis, i.e. : G91G00Z100., then Z axis does not move, and the remaining distance and program coordinates are displayed normally, but the machine coordinates are not changed. The system will synchronize the machine coordinates to the program coordinates until this function is turned off. If the function is enabled (C45 ON) during axial movement, the Z axis will not move and the remaining distance and program coordinates will be displayed normally, but the machine coordinates will stop when the function is activated. The system will synchronize the machine coordinates to the program coordinates until this function is turned off.
C046	Axis Coupling Request	The first set of axis coupling state switches, when Pr3825 is set to non-mechanical coupling, the actual coupling state is determined by this bit. When this FLAG is ON, the coupling axis enters the coupling state. When this FLAG is OFF, the coupling state is disabled. In the coupling state, when the master axis moves, the slave axis moves according to the set ratio.
C047	Inhibit Persist State Flush	When this FLAG is ON, the system is prohibited from storing persistent data. However, the system will force to synchronize once this FLAG turns from OFF to ON.
C048	Wait Function Skip Signal	All main system path skip the WAIT() command, and continues the look ahead in the following machining program.
C049	Macro Interruption Request	Interrupt signal of interrupted subprogram function. When this C Bit On, the program that was originally executed stops immediately and calls the interrupt subprogram.
C050	Over Travel X+	The X-axis machine hardware limit, when this FLAG ON controller will automatically enter Feedhold state. Meanwhile, it will not accept the JOG command or MPG command in this direction, and can be retracted only in the opposite direction.

No	Title	Description
C051	Over Travel X-	(As above)
C052	Over Travel Y+	(As above)
C053	Over Travel Y-	(As above)
C054	Over Travel Z+	(As above)
C055	Over Travel Z-	(As above)
C056	Over Travel C(6th)+	(As above)
C057	Over Travel C(6th)-	(As above)
C058		
C059		

No	Title	Description
C060	Spindle JOG Mode	When C60 ON, the spindle running speed will be forced to run at the spindle positioning drifting speed set by Pr1721. This function is usually used for milling machine spindle tool change, calibration or automatic lathe machine tending. C60 only affects and designate the speed of the spindle, which does not activate the spindle rotation; therefore, it still needs to be used with C64. The spindle is originally under C64, and follow S code command; under C64+C60, follow the Pr1721 command.

No	Title	Description
C061	Spindle Orientation Stop	When C61 is ON, the spindle positioning function is executed. This function must be equipped with position encoder on the spindle side or the gear ratio is 1:1. C61 can be disabled only when positioning completed; during positioning, press Reset (C37 ON) will not interrupt the positioning action. <u>General Spindle</u> - Pr1741 sets the speed of spindle positioning. After the spindle reaches the positioning speed, it starts to detect the signal. Pr1751 sets the spindle positioning offset angle. This angle is the offset angle in respect to the index signal. After replacing the motor, encoder, or belt, this parameter must be re-calibrated; Pr1761 sets the spindle positioning time; S61 is the spindle positioning completion signal. When C61 is ON, the spindle will stop first and then start positioning from stationary. - Pr1771 is invalid after version 10.116.26A, 10.116.36I. - Spindle positioning angle description: ? 未知的附加檔案 <u>General Serial Spindle</u> - Pr1791= 1~2, the positioning action is same as general spindle. - Pr1791= 3, the first positioning action is same as general spindle. After the second time, the positioning can be reached within one revolution by using the acceleration of Pr1801 / Pr1831 / Pr1841. The positioning angle is Pr1751. Since the whole process is position control, the controller can directly position the spindle to the set value even if the spindle is running at high speed. - When the positioning is completed, S61 is ON, and the spindle angle is displayed as the setting of "Pr1751" - spindle positioning angle description: same as the general spindle <u>Syntec Serial Spindle</u> - Use the built-in spindle quick positioning function in the driver. - The spindle positioning angle is determined by Pr1771. - Syntec M2 Pr1771 will be synchronized with the driver P6-20. The rules are as follows: - Power on, the P6-20 of the driver is synchronized to Pr1771~Pr1776 for the first time, - Afterwards, no matter modifying the controller parameters (Pr1771~Pr1776) or the driver parameters (P6-20), it will be synchronized in both sides. - Syntec M3 has no P6-20 parameters, so there will be no such synchronous action. - Pr1751 is invalid and must be set to 0, otherwise system issue SPD-026 alarm. - After positioning is completed, the spindle angle is displayed as 0 degrees. - Spindle positioning angle description : ? 未知的附加檔案 <u>Any point high speed positioning function</u>

No	Title	Description
		G10 L20 P0 R_ can be used to modify the positioning angle dynamically and display it synchronously at #1610. M19 C_ : The standard arbitrary point positioning function provided by the controller. It needs to register M0019 as Macro to use. (ex. Pr3601 set as 19). - The actual specifications of M0019 are as follows : - Read the C argument and set the spindle positioning angle through G10 L20 P0 R#3. (G10 L20 detailed usage please see the development manual) - As above, if there is no C argument, the default angle is set by Pr1751~Pr1756, that is, in a. #3 will be replaced by Pr1751~Pr1756. - After the positioning angle is determined, issue M19 to PLC for triggering C61 ON to perform positioning. (PLC needs to cooperate with C61 ON) - Support positioning when the spindle is rotating clockwise (counterclockwise), that is, directly position when C64 (C65) is ON. - The positioning speed and acceleration are controlled by the driver and are not affected by the controller parameter setting. - After positioning is executed, the positioning angle is restored to the angle set by the parameter. - For Syntec M2 spindle, the driver firmware must be updated to version 1.4.2 and above. <u>Operating restrictions</u> - In the tapping mode, this C Bit is invalid. - Because the controller will synchronize the coordinates when C63 is switched, it is recommended to use this function in the case of C63 OFF. Otherwise, the error coordinate synchronization may cause, and the machining action is not as expected after the positioning is completed. <u>Version compatibility description</u> - Refer to the page "Spindle Application - Spindle Positioning (C61)".
C062®	Skip Signal State	When C62 is ON, the measurement escape command G31 will be recording the current mechanical position and interrupt the G31 action, then jump directly to the next block to continue execution. If there are multiple G31 to be triggered in the program, it is recommended that C62 is upper triggered, preventing multiple G31 from being tripped within the same C62_ON.

No	Title	Description
C063	Spindle OFF	For non-inverter spindle type (Pr1791~≠0, 5, 6, 7), this C BIT ON means that the spindle is switched to the position control mode. Now controller will perform mechanical coordinate synchronization (motor feedback synchronize back to the mechanical coordinate); but when C BIT is OFF, it means to switch back to the spindle speed control mode, and its clockwise and counterclockwise rotation control is determined by C64/C65. <u>Compatibility change:</u> 1. Version 10.116 and before, when using M Code to switch the spindle C/S axis conversion, it is recommended that C38 in the PLC wait for S63 to ensure the correctness of the position control mode. 2. Version 9.x and before, the position control mode can be switched without using C63, but should follow the behavior of #1820 first turn on then turn off later to synchronize the coordinates, in order to ensure the correctness of the mechanical coordinates. <u>Operating restrictions:</u> In tapping mode, this C Bit is invalid.
C064	Rotating the spindle clockwise	For non-inverter spindle type (Pr1791~≠0, 5, 7), this signal is the spindle clockwise rotation command. In order to correctly use the spindle speed arrival check function Pr1991~, even if it is a variable frequency spindle, this C Bit must be triggered to notify the system spindle to enter the running state. <u>Operating restrictions:</u> In the tapping mode, this C Bit is invalid.
C065	Rotating the spindle counterclockwise	For non-inverter spindle type (Pr1791~≠0, 5, 7), this signal is the spindle counterclockwise rotation command. In order to correctly use the spindle speed arrival check function Pr1991~, even if it is a variable frequency spindle, this C Bit must be triggered to notify the system spindle to enter the running state. <u>Operating restrictions:</u> In the tapping mode, this C Bit is invalid.

No	Title	Description
C066®	X axis controlled by PLC	1. When this signal is ON, the PLC axis movement is executed. The movement command of the NC program does consider the movement of the PLC axis, that is, the coordinate synchronization between the NC axis and the PLC axis, but the movement speed of the PLC axis at this time will not be added to display on HMI. 2. When this signal is OFF, the PLC axis movement is executed. The movement command of the NC program does not consider the movement of the PLC axis, but the movement speed of the PLC axis will be added to display on HMI. 3. Following item 2, since this signal is OFF, the axis is not declared as PLC control. Therefore, when mode switches, the specific axis' PLC movement will be interrupted. 4. When NC program and the PLC axis simultaneously command the same axis, the axial command will have an additive effect, user should pay extra attention on this. 5. This signal can only be used in position control mode. If the axis is spindle, you need to switch to position control mode to perform PLC axis.
C067®	Y axis controlled by PLC	(As above)
C068®	Z axis controlled by PLC	(As above)
C069®	C axis(6th) controlled by PLC	(As above)
C070		
C071	Enable MPG No1	Under non-MPG mode, this signal is enabled (ON), and the input of the No. 1 MPG is used as an axial position command to move the corresponding axis. ※ Under MPG mode, the default MPG channel is enabled, so this signal is disabled (OFF), and the MPG signal cannot be blocked as the axial position command input. ※ The channel signal can be turned off by setting the parameter Pr2021 to 0.

No	Title	Description
C072	Enable MPG No2	(As above)
C073	Enable MPG No3	(As above)
C074		
C075		
C076		
C077		
C078		
C079	X HOME DOG	When HOME DOG signal is read from IO board, this C BIT needs to be ON, so need to add this C BIT in LADDER.
C080	Y HOME DOG	(As above)
C081	Z HOME DOG	(As above)
C082	C(6th) HOME DOG	(As above)

No	Title	Description
C083	The third software stroke limit switch	The third software stroke limit switch. Off: protection disabled; On: protection enabled. Please refer to parameters 2441~2480 for further explanation. (Version 10.116.x and after it is renamed to the third software stroke limit)
C084	Relieve the excessive amount of movement while power-off FLAG.	When the controller is powered off, and axial movement exceeds the setting range of Pr1021~1040 which R940 is triggered, this can be relieved by this C Bit.
C085		

No	Title	Description
C086	CHPST	Enabled Chopping function, relative parameter : Pr3951~3956, safety cautions as below; 1. Valid version: 10.118.19(Include) and before. 2. Cannot be mixed with the first path G81.1. 3. When C86 Off → On, it means the Chopping function is enabled. Chopping axis image will move to the reference point first, and then move back and forth between up and down vertex. 4. When C86 On → Off, it means the Chopping function is , and the Chopping axis will move to the lower vertex first and then back to the R point. 5. If the emergency stop (C36) is pushed during the Chopping process, the Chopping axis will stop immediately. 6. If the controller is reset (C37) during Chopping, the Chopping axis will move back to the specified R point coordinates before stopping. 7. When the controller has an alarm, the Chopping axis will act according to the alarm urgency level: a. The alarm that makes system enter Not Ready State (Servo Off), the Chopping axis will stop immediately. b. The alarm that stop the system from machining cycle, the Chopping axis will move back to the specified R point coordinates before stopping. c. The alarm does not stop the system from machining cycle, the Chopping axis will work normally. 8. Chopping cannot be used at the RTCP mode, otherwise COR-342 alarm is issued. 9. When Chopping is in progress, the Chopping axis can no longer input other movement commands, otherwise COR-339 alarm is issued. a. Movement commands such as: G0, G1, G2/G3, G31, G53, and etc. 10. When Chopping is in process, the Chopping axis cannot change the coordinate data, otherwise COR-340 alarm is issued. a. Coordinate data such as: G54 P1~G54 P100, G92/G92.1, G10 L2, G10 L1300, G68/G68.2/G68.3, #value(#1880~#1933, #20001~#20658), external coordinate offset, MPG offset. 11. When the Chopping axis is in the PLC axis execution state, or as spindle, or it has been specified by other path as the Chopping axis, COR-341 alarm will be issued. 12. Chopping axial does not support mirroring.

No	Title	Description
C087	CHLD	Chopping Feedhold function Valid version: 10.118.19 (include) and before. When C87 On → Off while in movement, the Chopping axis will stop and return to the R point; when C87 goes from Off → On, the Chopping axis action will activate again. While feedhold, you may change the upper and lower vertex's coordinates. There can be different actions when you change in different timings. Below are four possible situations: - When the upper vertex coordinates are modified, during the axial movement from the upper vertex to the lower vertex, the axis first moves to the lower vertex and then to the new upper vertex coordinates. When the axis moves to the lower vertex coordinates, the servo lag compensation function will also be canceled at the same time. When the new upper vertex is reached, the compensation will be activated again.  未知的附加檔案 - When the lower vertex coordinates are modified during the axial movement from the upper vertex to the lower vertex, the axis first moves to the upper vertex and then to the new lower vertex coordinates. When the axis moves to the upper vertex coordinates, the servo lag compensation function will also be canceled at the same time. When the new lower vertex is reached, the compensation will be activated again.  未知的附加檔案 - When the upper vertex coordinates are modified during the axial movement from the lower vertex to the upper vertex, the axis moves to the upper vertex first, then to the lower vertex and the new upper vertex coordinates. When the axis moves to the lower vertex coordinates, the servo lag compensation function will also be canceled at the same time. When the new upper vertex is reached, the compensation will be activated again.  未知的附加檔案 - When the lower vertex coordinates are modified during the axial movement from the lower vertex to the upper vertex, the axis moves directly to the new lower vertex coordinates after moving to the upper vertex coordinates. When the axis moves to the upper vertex coordinates, the servo lag compensation function will also be canceled at the same time. When the new lower vertex is reached, the compensation will be activated again.  未知的附加檔案
C088	Disable Screen Saver	Disable the screen saver after screen saver is already activated by on this bit. The recommended igniting time is 500ms at least.(Only available on CE system) Valid version: 10.114.53, 10.115.55, 10.114.50C

No	Title	Description
C089	Edit inhibition	Prohibit edit program mode (Including MDI edit) 0: Allow program editing 1: Prohibit program edition.
C090	Program Edit Screen	Switch to program edit screen.
C091	Graph Dialog Screen	Switch to graphical dialog screen.
C092	Monitor Screen	Switch to program processing monitor screen.
C093	Position Screen	Switch to Position Monitor screen
C094	Tool Offset Screen	Switch to Tool Offset screen
C095		
C096	File Management Screen	Switch to File Management screen
C097		
C098		
C099		
C100		

No	Title	Description
C101 ~C132	Macro mapping variable	Mapping to the macro system variables (#6001~#6032)
C133	Axis Exchange Request 1	First path switch function Enable axis exchange status of two axis assigned by Pr3721 & Pr3722.
C134	Axis Exchange Request 2	Second path switch function Enable axis exchange of two axis assigned by Pr3726 & Pr3727.
C135	Axis Exchange Request 3	Third path switch function Enable axis exchange of two axis assigned by Pr3731 & Pr3732.
C136	Axis Exchange Request 4	Fourth path switch function Enable axis exchange of two axis assigned by Pr3736 & Pr3737.
C137	Tension Control Switch	Two axis designated by Pr2901、 Pr2902 will enter tension control state.
C138		
C139		

No	Title	Description
C140	4 th Axis OT +	The 4 th axis machine positive hardware limit, when this FLAG ON controller will automatically enters Feedhold state. Meanwhile, it will not accept the JOG command or MPG command in this direction, and can be retracted only in the opposite direction.
C141	4 th Axis OT -	(As above)
C142	5 th Axis OT +	(As above)
C143	5 th Axis OT -	(As above)
C144	7 th Axis OT +	(As above)
C145	7 th Axis OT -	(As above)
C146	8 th Axis OT +	(As above)
C147	8 th Axis OT -	(As above)
C148	9 th Axis OT +	(As above)
C149	9 th Axis OT -	(As above)
C150	10 th Axis OT +	(As above)

No	Title	Description
C151	10 th Axis OT -	(As above)
C152	11 th Axis OT +	(As above)
C153	11 th Axis OT -	(As above)
C154	12 th Axis OT +	(As above)
C155	12 th Axis OT -	(As above)
C156	13 th Axis OT +	(As above)
C157	13 th Axis OT -	(As above)
C158	14 th Axis OT +	(As above)
C159	14 th Axis OT -	(As above)
C160	15 th Axis OT +	(As above)
C161	15 th Axis OT -	(As above)

No	Title	Description
C162	16 th Axis OT +	(As above)
C163	16 th Axis OT -	(As above)
C164	17th Axis OT +	(As above)
C165	17th Axis OT -	(As above)
C166	18th Axis OT +	(As above)
C167	18th Axis OT -	(As above)
C168		
C169		
C170®	4th Axis JOG +	User JOG MODE, press the 4th Axis JOG+ button on the panel. When this FLAG is ON in the LADDER, controller will move along this direction.
C171®	4th Axis JOG -	(As above)
C172®	5th Axis JOG +	(As above)

No	Title	Description
C173®	5th Axis JOG -	(As above)
C174®	7th Axis JOG +	(As above)
C175®	7th Axis JOG -	(As above)
C176®	8th Axis JOG +	(As above)
C177®	8th Axis JOG -	(As above)
C178®	9th Axis JOG +	(As above)
C179®	9 th JOG -	(As above)
C180®	10 th Axis JOG +	(As above)
C181®	10 th Axis JOG -	(As above)
C182®	11 th Axis JOG +	(As above)
C183®	11 th Axis JOG -	(As above)
C184®	12 th Axis JOG +	(As above)

No	Title	Description
C185®	12 th Axis JOG -	(As above)
C186®	13 th Axis JOG +	(As above)
C187®	13 th Axis JOG -	(As above)
C188®	14 th Axis JOG +	(As above)
C189®	14 th Axis JOG -	(As above)
C190®	15 th Axis JOG +	(As above)
C191®	15 th Axis JOG -	(As above)
C192®	16 th Axis JOG +	(As above)
C193®	16 th Axis JOG -	(As above)
C194	17th Axis JOG +	(As above)
C195	17th Axis JOG -	(As above)

No	Title	Description
C196	18th Axis JOG +	(As above)
C197	18th Axis JOG -	(As above)
C198		
C199		
C200	4th Axis Home Dog	When HOME DOG is read from IO board, LADDER needs to trigger this C BIT.
C201	5th Axis Home Dog	(As above)
C202	7th Axis Home Dog	(As above)
C203	8th Axis Home Dog	(As above)
C204	9th Axis Home Dog	(As above)
C205	10 th Axis Home Dog	(As above)

No	Title	Description
C206	11 th Axis Home Dog	(As above)
C207	12 th Axis Home Dog	(As above)
C208	13 th Axis Home Dog	(As above)
C209	14 th Axis Home Dog	(As above)
C210	15 th Axis Home Dog	(As above)
C211	16 th Axis Home Dog	(As above)
C212	17 th Axis Home Dog	(As above)
C213	18th Axis Home Dog	(As above)
C214		

No	Title	Description
C215	4th Axis MPG Selection	The 4th axis' axial MPG selection. When this FLAG is ON, controller will input pulse of the MPG and drive the 4th axis to move in relative displacement.
C216	5th Axis MPG Selection	(As above)
C217	7th Axis MPG Selection	(As above)
C218	8th Axis MPG Selection	(As above)
C219	9 th Axis MPG Selection	(As above)
C220	10 th Axis MPG Selection	(As above)
C221	11 th Axis MPG Selection	(As above)
C222	12 th Axis MPG Selection	(As above)

No	Title	Description
C223	13 th Axis MPG Selectio n	(As above)
C224	14 th Axis MPG Selectio n	(As above)
C225	15 th Axis MPG Selectio n	(As above)
C226	16 th Axis MPG Selectio n	(As above)
C227	17 th Axis MPG Selectio n	(As above)
C228	18 th Axis MPG Selectio n	(As above)
C229		
C230	Set 4 th Axis Ma chine Position	Set the axial machine coordinates, the content filled is determined by R38.

No	Title	Description
C231	Set 5 th Axis Ma chine Position	(As above)
C232	Set 7 th Axis Ma chine Position	(As above)
C233	Set 8 th Axis Ma chine Position	(As above)
C234	Set 9 th Axis Ma chine Position	(As above)
C235	Set 10 th Axis Ma chine Position	(As above)
C236	Set 11 th Axis Ma chine Position	(As above)
C237	Set 12 th Axis Ma chine Position	(As above)
C238	Set 13 th Axis Ma chine Position	(As above)

No	Title	Description
C239	Set 14 th Axis Machine Position	(As above)
C240	Set 15 th Axis Machine Position	(As above)
C241	Set 16 th Axis Machine Position	(As above)
C242	Set 17 th Axis Machine Position	(As above)
C243	Set 18 th Axis Machine Position	(As above)
C244		
C245	4 th Axis MANUAL CONTROL	When axial trips from control mode, and enters the feedback monitoring mode. Now controller can no longer send commands to the axis. If the axis moves due to external force, the coordinates on MHI will also be updated. <u>Remark:</u> When the axis is switched back to the control mode from the feedback monitoring mode, if the machine is still moving, will cause zero speed check fails (Pr901~), the system will issue a MOT-020/MOT-021 alarm, and the axis needs to redo homing.

No	Title	Description
C246	5 th Axis MANUA L CONTR OL	(As above)
C247	7 th Axis MANUA L CONTR OL	(As above)
C248	8 th Axis MANUA L CONTR OL	(As above)
C249	9 th Axis MANUA L CONTR OL	(As above)
C250	10 th Axis MANUA L CONTR OL	(As above)
C251	11 th Axis MANUA L CONTR OL	(As above)
C252	12 th Axis MANUA L CONTR OL	(As above)

No	Title	Description
C253	13 th Axis MANUA L CONTR OL	(As above)
C254	14 th Axis MANUA L CONTR OL	(As above)
C255	15 th Axis MANUA L CONTR OL	(As above)
C256	16 th Axis MANUA L CONTR OL	(As above)
C257	17 th Axis MANUA L CONTR OL	(As above)
C258	18 th Axis MANUA L CONTR OL	(As above)
C259		

No	Title	Description
C260®	4 th axis controlled by PLC	1. When this signal is ON, the PLC axis movement is executed. The movement command of the NC program does consider the movement of the PLC axis, that is, the coordinate synchronization between the NC axis and the PLC axis, but the movement speed of the PLC axis at this time will not be added to display on HMI. 2. When this signal is OFF, the PLC axis movement is executed. The movement command of the NC program does not consider the movement of the PLC axis, but the movement speed of the PLC axis will be added to display on HMI. 3. Following item 2, since this signal is OFF, the axis is not declared as PLC control. Therefore, when mode switches, the specific axis' PLC movement will be interrupted. 4. When NC program and the PLC axis simultaneously command the same axis, the axial command will have an additive effect, user should pay extra attention on this. 5. This signal can only be used in position control mode. If the axis is spindle, you need to switch to position control mode to perform PLC axis.
C261®	5 th axis controlled by PLC	(As above)
C262®	7 th axis controlled by PLC	(As above)
C263®	8 th axis controlled by PLC	(As above)
C264®	9 th axis controlled by PLC	(As above)
C265®	10 th axis controlled by PLC	(As above)

No	Title	Description
C266®	11 th axis controll ed by PLC	(As above)
C267®	12 th axis controll ed by PLC	(As above)
C268®	13 th axis controll ed by PLC	(As above)
C269®	14 th axis controll ed by PLC	(As above)
C270®	15 th axis controll ed by PLC	(As above)
C271®	16 th axis controll ed by PLC	(As above)
C272	17 th axis controll ed by PLC	(As above)
C273	18 th axis controll ed by PLC	(As above)

No	Title	Description
C274	Enable MPG No4	Under non-MPG mode, this signal is enabled (ON), and the input of the No. 4 MPG is used as an axial position command to move the corresponding axis. ※ Under MPG mode, the default MPG channel is enabled, so this signal is disabled (OFF), and the MPG signal cannot be blocked as the axial position command input. ※ The channel signal can be turned off by setting the parameter Pr2024 to 0.
C275	Enable MPG No5	(As above)
C276	Enable MPG No6	(As above)
C277	Enable MPG No7	(As above)
C278		
C279		
C280		
C281		
C282		
C283		
C284		
C285		

No	Title	Description
C286		
C287		
C288		
C289		
C290		
C291		
C292		
C293		
C294		
C295		
C296		
C297		
C298		
C299		
C300	Reload ECam	The Reload ECam insertion point is modified in the R value. (For electronic cam function)

No	Title	Description
C301	Enable 1 st ECam	Enable the first group of electronic cams. When this Bit ON, the first group of electronic cam will start executing, and the S Bit with same number will be ON; when this Bit is OFF, the ECam function will only be terminated until all curve paths are finished.
C302	Enable 2 nd ECam	(As above)
C303	Enable 3 rd ECam	(As above)
C304	Enable 4 th ECam	(As above)
C305	Enable 5 th ECam	(As above)
C306	Enable 6 th ECam	(As above)
C307	Enable 7 th ECam	(As above)
C308	Enable 8 th ECam	(As above)
C309		
C310		

No	Title	Description
C311		
C312		
C313		
C314		
C315		
C316		
C317 ~C319		
C320 ~C350	For robot kernel	Robot kernel usage area (not customizable area)
C351 ~C400		
C401 ~C488	Standard keyboard Scan Code	Mapping to the standard PC keyboard scan code 1~88
C489	OEM extended key	Application self-defined extension keys 89-111 are user defined by individual applications. Mapping to PC keyboard ALT-1
C490		Mapping to PC keyboard ALT-2

No	Title	Description
C491		Mapping to PC keyboard ALT-3
C492		Mapping to PC keyboard ALT-4
C493		Mapping to PC keyboard ALT-5
C494		Mapping to PC keyboard ALT-6
C495		Mapping to PC keyboard ALT-7
C496		Mapping to PC keyboard ALT-8
C497		Mapping to PC keyboard ALT-9
C498		Mapping to PC keyboard ALT-0
C499		Mapping to PC keyboard ALT-B
C500		Mapping to PC keyboard ALT-V
C501		Mapping to PC keyboard ALT-E
C502		Mapping to PC keyboard ALT-F
C503		Mapping to PC keyboard ALT-I
C504		Mapping to PC keyboard ALT-J
C505		Mapping to PC keyboard ALT-K
C506		Mapping to PC keyboard ALT-L

No	Title	Description
C507		Mapping to PC keyboard ALT-W
C508		Mapping to PC keyboard ALT-N
C509		Mapping to PC keyboard ALT-S
C510		Mapping to PC keyboard ALT-T
C511		Mapping to PC keyboard ALT-U

All the C bits will be read simultaneously and only the C bit with ® mark is processed at real time. Other signals will be send to the stack after being read, then run by sequence non-simultaneously.

1.2 S Bit Interface (From CNC To PLC)

No	Title	Description
S000	Cycle Start Light	During machining (BUSY status), this FLAG will be ON.
S001	Feed Hold Light	Pause Machining (FEEDHOLD status), this FLAG will be ON.
S002	Block stop	During single block stop (BLOCK STOP status), this FLAG will be ON.
S003	Feedhold inhibit	When this FLAG is ON, it indicates that feedhold status is prohibited now.
S004	Rigid Tapping	First spindle is on rigid tapping mode.
S005		

No	Title	Description
S006	X Axis Busy	When this FLAG is ON, it indicates that the X-axis manual function (MPG/JOG/Homing/PLC axis) is being executed, and the new manual command cannot be accepted. When this FLAG is OFF, the X-axis is in the Idle state and is allowed to accept new manual command.
S007	Y Axis Busy	(As above)
S008	Z Axis Busy	(As above)
S009	C(6 th) Axis Busy	(As above)
S010		
S011	2 nd of Multiple M code read	When Pr3810 (activating the same block M code simultaneous execution function) is set to 1, and the controller executes the second M code in the same block, this FLAG will be ON, and inform LADDER to execute M code, the content of M code is stored in R2051. After the controller issues this FLAG, it will wait until the C BIT FLAG ON of M code FIN, then starting to execute the next BLOCK.
S012	3 rd of Multiple M code read	(As above, and M mode content is stored in R2052)
S013	4 th of Multiple M code read	(As above, and M mode content is stored in R2053)
S014	5 th of Multiple M code read	(As above, and M mode content is stored in R2054)
S015		

No	Title	Description
S016	X Axis Home OK	X Homing OK, X axis HOME process finished, and this FLAG will be ON. The software stroke limit protection will be activated after homing completed. Therefore, in the case that this S BIT FLAG is not ON, LADDER should write not allowed to cycle start and give homing not complete alarm.
S017	Y Axis Home OK	(As above)
S018	Z Axis Home OK	(As above)
S019	C(6 th) Axis Home OK	(As above)
S020		
S021	During Tool Withdraw	After entering the tool retract function and leaving the original machining path, this FLAG will be ON; until the tool returned and repositioned to the original machining program path, and this FLAG will be OFF.
S022		
S023	Servo Not Ready	- Description - When all the axes(except axes in spindle mode or watch mode) of CNC system are in servo-on state, this S bit will be off. - When NOT all the axes(except axes in spindle mode or watch mode) of CNC system are in servo-on state, this S bit will be on. - Attention note - Valid version: 10.118.24A, 10.118.25A, 10.118.28N, 10.118.37 and later. - In 10.118.40 and later version, when Pr741 is set to 1, this status is not including the axis of axis group of independent operation.
S024		
S025		

No	Title	Description
S026		
S027		
S028		
S029	M Code Read	When CNC executes M CODE, this FLAG will be ON, and tell LADDER to execute M CODE. The content of M CODE is saved in R1. After CNC issues this FLAG, it has to wait until M CODE FIN's C BIT FLAG ON, then executes the next BLOCK.
S030	Distribution End	After MOTION G executed a BLOCK, it sends a DEN signal, usually used when G and M CODE have simultaneous action.
S031	Alarm	- When an alarm appears in CNC or driver, and makes the system must stop or feedhold, and this flag will be ON. - In 10.118.40 and later version, when Pr741 is set to 1, this flag is not including the axis group of independent operation has alarm. Controller alarm: CNC: COR alarm, MOT alarm, SPD alarm, OP alarm, COM alarm; ROT: ROT alarm; SPLC: SERIALPLCAXIS alarm; SRI: SRI alarm; Laser marking: LaserCtrl alarm.
S032	Reset	- When press RESET on the panel, this FLAG will send a ONE SHOT signal. - When _Reset Action is triggered, this FLAG will send a ONE SHOT signal.
S033	NC Ready	- After the boot is completed and the system is normal, NC READY FLAG will be ON. - In 10.118.40 and later version, when Pr741 is set to 1, this status is nothing to do with the axis group of independent operation.
S034	Require Part Count Reached	When the required workpieces number is reached, this S FLAG will be ON, and when Cycle Start or Reset, this FLAG will be OFF.
S035	The Part Count M-code in Main Program	When main program reads a part count M code (default as M99), this FLAG sends out a ON/OFF one shot signal. Relative information: Parameter3804, C39

No	Title	Description
S036	In Guidance Mode	When CNC is under Guidance Function mode, this BIT will be ON.
S037	Loss data	When the user data (Registry.dat) is lost, this Bit will be ON.
S038	MACRO WAIT	When system enters waiting mode (MACRO function WAIT() is read)
S039		
S040		
S041	PLC Alarm	When the PLC has an alarm, this flag will be ON, and when the PLC alarm disappears, the flag will be OFF. The first 16 bits (R4x.0~R4x.15) of R40~R49 are PLC alarm messages, and the PLC warning message after 16th bits will not trigger S41. (The 0~15 bits of R40~R49 are PLC Alarms, and 16~31 bits of them are PLC Warnings.)
S042	CNC Warning	- When the controller or driver has an warning, this flag will be ON, and when all Warnings disappear, the flag will be OFF. - In 10.118.40 and later version, when Pr741 is set to 1, this flag is not including the axis group of independent operation has warning.
S043	PLC Warning	When PLC has an alarm, this flag ON, and when all alarms disappear, the flag OFF.
S044		
S045		
S046		
S047		
S048		

No	Title	Description
S049		
S050		
S051		
S052		
S053	B Code Read	This function is enabled by Pr3806. When CNC executes to B CODE, this FLAG will be ON, and LADDER will be told to execute B CODE, the content of B CODE will be placed in R5. After the CNC sends this FLAG, it will wait until the B CODE FINISH (C38)'s C BIT FLAG ON, then start executing the next BLOCK. Only the first MST channel supports thePLC B CODE execution (refer to Pr721~Pr730)
S054	S Code Read	When CNC executes to S CODE, this FLAG will be ON, and LADDER will be told to execute S CODE, the content of S CODE will be placed in R2. After CNC issues this FLAG, it will wait until the S CODE FINISH (C38)'s C BIT FLAG ON, then start executing the next BLOCK.
S055		
S056		
S057		
S058		

No	Title	Description						
S059	Spindle Threading Speed Arrival	Spindle Speed Arrival when Threading.						
		<table><tr><th>Spindle CW/CCW Signal (C64/C65, R583.x/R584.x)</th><th>Spindle Speed Arrival Condition</th></tr><tr><td>On</td><td>The speed difference of spindle feedback Fbk and spindle command Cmd is less than the command speed Cmd multiplies a speed arrival check range percentage (Pr1992 or Pr1993). i.e. $Fdbk - Cmd < Cmd * Pr1993 * 0.01$</td></tr><tr><td>Off</td><td>Spindle feedback less than 50 RPM. i.e. $Fdbk < 50 \text{ (RPM)}$</td></tr></table>	Spindle CW/CCW Signal (C64/C65, R583.x/R584.x)	Spindle Speed Arrival Condition	On	The speed difference of spindle feedback Fbk and spindle command Cmd is less than the command speed Cmd multiplies a speed arrival check range percentage (Pr1992 or Pr1993). i.e. $ Fdbk - Cmd < Cmd * Pr1993 * 0.01$	Off	Spindle feedback less than 50 RPM. i.e. $Fdbk < 50 \text{ (RPM)}$
		Spindle CW/CCW Signal (C64/C65, R583.x/R584.x)	Spindle Speed Arrival Condition					
		On	The speed difference of spindle feedback Fbk and spindle command Cmd is less than the command speed Cmd multiplies a speed arrival check range percentage (Pr1992 or Pr1993). i.e. $ Fdbk - Cmd < Cmd * Pr1993 * 0.01$					
		Off	Spindle feedback less than 50 RPM. i.e. $Fdbk < 50 \text{ (RPM)}$					
		When the spindle is not equipped with an encoder (Pr1711~ = 0) , the spindle speed arrival determined with following conditions.						
		<table><tr><th>Spindle Type</th><th>Spindle Speed Arrival Condition</th></tr><tr><td>Serial Bus Spindle</td><td>Directly reflects the spindle speed arrival information provided by driver. (In serial communication, it can be known from the V_CMP bit in the status of the driver packet, S59 reacts to this bit, and the PLC can be used.)</td></tr><tr><td>Pulse Train Spindle</td><td>Controller side considers spindle speed arrived S59 On. (CNC's S59 is always On) If you want to check the spindle speed arrival, you can send an O bit to the TB16IN through the driver, PLC can get this information, and check the speed arrival in PLC.</td></tr></table>	Spindle Type	Spindle Speed Arrival Condition	Serial Bus Spindle	Directly reflects the spindle speed arrival information provided by driver. (In serial communication, it can be known from the V_CMP bit in the status of the driver packet, S59 reacts to this bit, and the PLC can be used.)	Pulse Train Spindle	Controller side considers spindle speed arrived S59 On. (CNC's S59 is always On) If you want to check the spindle speed arrival, you can send an O bit to the TB16IN through the driver, PLC can get this information, and check the speed arrival in PLC.
		Spindle Type	Spindle Speed Arrival Condition					
		Serial Bus Spindle	Directly reflects the spindle speed arrival information provided by driver. (In serial communication, it can be known from the V_CMP bit in the status of the driver packet, S59 reacts to this bit, and the PLC can be used.)					
		Pulse Train Spindle	Controller side considers spindle speed arrived S59 On. (CNC's S59 is always On) If you want to check the spindle speed arrival, you can send an O bit to the TB16IN through the driver, PLC can get this information, and check the speed arrival in PLC.					
S060	1st Set Spindle Superimposition Finish	1st set spindle superimposition completion signal.						

No	Title	Description
S061	Spindle Orientation Stop Finish	Spindle positioning completion signal
S062	1st Set Spindle Synchronization Finish	1st set spindle speed/angle synchronization is completed.
S063	Spindle axis mode	When this S BIT is ON, it indicates that the spindle is in the position control mode; when this S BIT is OFF, it indicates that the spindle is in the spindle speed control mode.

No	Title	Description						
S064	Spindle Cutting Speed Arrival	Spindle speed arrival signal while cutting.						
		<table><tr><th>Spindle CW/CCW Signal: C64/C65, R583.x/R584.x</th><th>Spindle Speed Arrival Criteria</th></tr><tr><td>On</td><td>The speed difference of spindle feedback Fbk and spindle command Cmd is less than the command speed Cmd multiplies a speed arrival check range percentage (Pr1992 or Pr1993). i.e. $Fbk - Cmd < Cmd * Pr1992 * 0.01$</td></tr><tr><td>Off</td><td>Spindle speed feedback < 50RPM i.e. $Fbk < 50 \text{ (RPM)}$</td></tr></table>	Spindle CW/CCW Signal: C64/C65, R583.x/R584.x	Spindle Speed Arrival Criteria	On	The speed difference of spindle feedback Fbk and spindle command Cmd is less than the command speed Cmd multiplies a speed arrival check range percentage (Pr1992 or Pr1993). i.e. $ Fbk - Cmd < Cmd * Pr1992 * 0.01$	Off	Spindle speed feedback < 50RPM i.e. $Fbk < 50 \text{ (RPM)}$
		Spindle CW/CCW Signal: C64/C65, R583.x/R584.x	Spindle Speed Arrival Criteria					
		On	The speed difference of spindle feedback Fbk and spindle command Cmd is less than the command speed Cmd multiplies a speed arrival check range percentage (Pr1992 or Pr1993). i.e. $ Fbk - Cmd < Cmd * Pr1992 * 0.01$					
		Off	Spindle speed feedback < 50RPM i.e. $Fbk < 50 \text{ (RPM)}$					
		When the spindle is not equipped with an encoder (Pr1711~ = 0) , the spindle speed arrival determined with following conditions.						
		<table><tr><th>Spindle Type</th><th>Spindle Speed Arrival Criteria</th></tr><tr><td>Serial Bus Spindle</td><td>Directly reflects the spindle speed arrival information provided by driver. (In serial communication, it can be known from the V_CMP bit in the status of the driver packet, S59 reacts to this bit, and the PLC can be used.)</td></tr><tr><td>Pulse Train Spindle</td><td>Controller side considers spindle speed arrived S64 On. (CNC's S64 is always On) If you want to check the spindle speed arrival, you can send an O bit to the TB16IN through the driver, PLC can get this information, and check the speed arrival in PLC.</td></tr></table>	Spindle Type	Spindle Speed Arrival Criteria	Serial Bus Spindle	Directly reflects the spindle speed arrival information provided by driver. (In serial communication, it can be known from the V_CMP bit in the status of the driver packet, S59 reacts to this bit, and the PLC can be used.)	Pulse Train Spindle	Controller side considers spindle speed arrived S64 On. (CNC's S64 is always On) If you want to check the spindle speed arrival, you can send an O bit to the TB16IN through the driver, PLC can get this information, and check the speed arrival in PLC.
		Spindle Type	Spindle Speed Arrival Criteria					
		Serial Bus Spindle	Directly reflects the spindle speed arrival information provided by driver. (In serial communication, it can be known from the V_CMP bit in the status of the driver packet, S59 reacts to this bit, and the PLC can be used.)					
		Pulse Train Spindle	Controller side considers spindle speed arrived S64 On. (CNC's S64 is always On) If you want to check the spindle speed arrival, you can send an O bit to the TB16IN through the driver, PLC can get this information, and check the speed arrival in PLC.					
S065								

No	Title	Description
S066		
S067		
S068		
S069	T Code Read	When the controller executes to T CODE, this FLAG will be ON, and LADDER will be told to execute T CODE, the content of T CODE will be placed in R3. After the controller issues this FLAG, it will wait until the T CODE FIN's C BIT FLAG ON, then start executing the next BLOCK.
S070	At start point	When this signal is ON, it indicates that the process is at the start of a program. After run into the program, S70 will OFF. And S70 will be ON after RESET(EX: M30).
S071	At end point	When this signal is ON, it indicates that the process is at the end of a program. S71 will be OFF when program is running. Notice that S71 will stay ON, before the program is start again. S71 will be ON after RESET.
S072	2nd Set Spindle Synchronization Finish	2nd set spindle speed/angle synchronization is completed.
S073	3rd Set Spindle Synchronization Finish	3rd set spindle speed/angle synchronization is completed.
S074		
S075		
S076		

No	Title	Description
S077		
S078		
S079		
S080	M00 Signal	M00 is read flag. This flag is automatically enabled after the M code is read until the system reset (Reset) or M Code Finish (C38 On). Note: 1. Valid version: 10.114.56U, 10.116.0R, 10.116.10C, 10.116.13 2. Only support the first MST channel. 3. To ensure that the system properly enables this flag, it is recommended that in PLC applications the C38 is off until the kernel enables this flag.
S081	M01 Signal	M01 is read flag. When the optional stop function (C44) is activated, this flag will be automatically enabled after the M code is read until the system reset (Reset) or M Code Finish (C38 On). Note: Same as S080.
S082	M02 Signal	M02 is read flag. this flag will be automatically enabled after the M code is read until the system reset (Reset) or M Code Finish (C38 On). Note: Same as S080.
S083	M30 Signal	M30 is read flag. this flag will be automatically enabled after the M code is read until the system reset (Reset) or M Code Finish (C38 On). Note1: Same as S080. Note2: If below conditions both fulfilled, M30 will enable the program return function (Valid version: 10.116.41, 10.117.41 and after). M30 will not be placed in R1 for LADDER to read, and S083 is not supported at this time. 1. Pr732 * Multi-program MST channel, set to 1 2. C015 ON
S084		
S085		

No	Title	Description
S086		
S087		
S088	Screen Saver	Screen saver mode enabled. (Only in CE system)
S089		
S090		
S091		
S092		
S093		
S094		
S095		
S096		
S097	Exe File CRC error	When this signal is ON, it indicates that system's specific file encounters CRC check error. This condition usually occurs. It may be that the system is infected with the virus and cannot be recovered. The relevant records will be stored in C:\CRC_Chk.log. <u>Compatibility change:</u> Valid version starts from 10.114.52.
S098	OCAPI Server is Running	Whether OCAPI Server is activated. When OCAPI Server is activated, this Bit is set to True; and when OCAPI Server is deactivated, this Bit is set to False.

No	Title	Description
S099	Dipole connection in use	Whether Dipole connection is in progress. When any remote program is connected to the kernel, this Bit is set to 1; and when no remote program is connected, this Bit is set to False.
S100	DOS USB in use	on: DOS USB in use; off: DOS USB not in use.
S101~S132		Mapping macro system variable (#6001~#6032)
S133	Axis Exchange Status 1	First set of axis exchange successfully flag. The axis exchange is completed inside the system, and this state changes.
S134	Axis Exchange Status 2	Second set of axis exchange successfully flag. The axis exchange is completed inside the system, and this state changes.
S135	Axis Exchange Status 3	Third set of axis exchange successfully flag. The axis exchange is completed inside the system, and this state changes.
S136	Axis Exchange Status 4	Fourth set of axis exchange successfully flag. The axis exchange is completed inside the system, and this state changes.
S137		
S138		
S139		
S140	4 th Axis Home OK	4th axis Homing OK, 4th axis axis HOME process finished, and this FLAG will be ON. The software stroke limit protection will be activated after homing completed. Therefore, in the case that this S BIT FLAG is not ON, LADDER should write not allowed to cycle start and give homing not complete alarm.

No	Title	Description
S141	5 th Axis Home OK	(As above)
S142	7 th Axis Home OK	(As above)
S143	8 th Axis Home OK	(As above)
S144	9 th Axis Home OK	(As above)
S145	10 th Axis Home OK	(As above)
S146	11 th Axis Home OK	(As above)
S147	12 th Axis Home OK	(As above)
S148	13 th Axis Home OK	(As above)
S149	14 th Axis Home OK	(As above)
S150	15 th Axis Home OK	(As above)
S151	16 th Axis Home OK	(As above)

No	Title	Description
S152	17 th Axis Home OK	(As above)
S153	18 th Axis Home OK	(As above)
S154		
S155	4 th Axis Busy	When this FLAG is ON, it indicates that the X axis is being executed, and a new command cannot be accepted. When this FLAG is OFF, the X axis is in the Idle state, allowing new commands to be accepted.
S156	5 th Axis Busy	(As above)
S157	7 th Axis Busy	(As above)
S158	8 th Axis Busy	(As above)
S159	9 th Axis Busy	(As above)
S160	10 th Axis Busy	(As above)
S161	11 th Axis Busy	(As above)
S162	12 th Axis Busy	(As above)
S163	13 th Axis Busy	(As above)

No	Title	Description
S164	14 th Axis Busy	(As above)
S165	15 th Axis Busy	(As above)
S166	16 th Axis Busy	(As above)
S167	17 th Axis Busy	(As above)
S168	18 th Axis Busy	(As above)
S169		
S170		

No	Title	Description																		
S171~ S186	Servo Alarm State	Alarm state for the port of motion card. Pulse Train Drives: This S bit will be activated when motion card does not receive alarm signal. Serial Bus Drives: This S bit will be activated when Servo Drive sens out a alarm signal. system reserved S171: Pulse Train Drive- Alarm state for the port 1; Serial Bus Drive- The station number 1 drive. S172: Pulse Train Drive- Alarm state for the port 2; Serial Bus Drive- The station number 2 drive. ... S186: Pulse Train Drive- Alarm state for the port 16; Serial Bus Drive- The station number 16 drive. Note : This function does not support Servo6. Operation Restriction : 1. The slot chassis type Servo6 axis card does not support this feature. 2. This function is only supported by pulse train type and M2 type. 3. Following from item 2, when using other communication types, please use S31 and S42. Pulse Train type : Indicates the signal status of the servo alarm contact This S-Bit only reflects the status of the servo alarm contact corresponding to the axis card port. When there is signal input, this S-Bit OFF; when there is no signal input, this S-Bit ON. According to the parameter "Pr10 servo alarm contact type", the system will trigger the corresponding axial "MOT-9 driver alarm" when this S-Bit ON/OFF is triggered. <table><tr><th>Servo Alarm Contact Type</th><th>Port Alarm Signal</th><th>S-17x</th><th>System Alarm</th></tr><tr><td rowspan="2">Pr10 = 0 (A contact)</td><td>No</td><td>ON</td><td>No</td></tr><tr><td>Yes</td><td>OFF</td><td>MOT-9</td></tr><tr><td rowspan="2">Pr10 = 1 (B contact)</td><td>No</td><td>ON</td><td>MOT-9</td></tr><tr><td>Yes</td><td>OFF</td><td>No</td></tr></table>	Servo Alarm Contact Type	Port Alarm Signal	S-17x	System Alarm	Pr10 = 0 (A contact)	No	ON	No	Yes	OFF	MOT-9	Pr10 = 1 (B contact)	No	ON	MOT-9	Yes	OFF	No
Servo Alarm Contact Type	Port Alarm Signal	S-17x	System Alarm																	
Pr10 = 0 (A contact)	No	ON	No																	
	Yes	OFF	MOT-9																	
Pr10 = 1 (B contact)	No	ON	MOT-9																	
	Yes	OFF	No																	

No	Title	Description
		S171 : Corresponding to the 1st axis card port. S172 : Corresponding to the 2nd axis card port ... S186 : Corresponding to the 16th axis card port M2 Serial : Indicates the driver alarm state If an M2 serial driver is used, the corresponding S-bit On will occur when a driver alarm occurs. S171 : Corresponding to the driver station no. which is set to be 1. S172 : Corresponding to the driver station no. which is set to be 2. ... S186 : Corresponding to the driver station no. which is set to be 16. i.e. : In the M2 serial environment, when the driver station number which set as 3 gives alarm, S173 will be On.
S190~ S299		(System reserved area)
S300	ECam Reload Status	Reload the ECam insertion point R value. When this Bit is On, it means that reload is completed and the electronic cam can be activated; when this Bit is Off, Reloading.
S301	1 st ECam Busy	The first group of electronic cams is busy. When this Bit is On, it means that the first group of electronic cams are working; when this Bit Off, the cam relationship motion between the master and slave axis is canceled.
S302	2 nd ECam Busy	(As above) When this Bit On is on, it indicates that the first group of electronic cams are in use; when this Bit is off, the master and slave axis are not working according to cam relationship.
S303	3 rd ECam Busy	(As above)
S304	4 th ECam Busy	(As above)
S305	5 th ECam Busy	(As above)

No	Title	Description
S306	6 th ECam Busy	(As above)
S307	7 th ECam Busy	(As above)
S308	8 th ECam Busy	(As above)
S309~ S319		
S321~ S250	For robot kernel	Robot kernel usage area (non-product customization area).
S351~ S400	Keyboard Sbit	S Bit mapping to keyboard scan code, and now only S355~S374 are open.
S401~ 488	Keyboard Scan Code	Mapping to the keyboard scan code 1~88.
~S511	Reserved for CNC	System reserved area.

1.3 PC Keyboard C/S Mapping

1.3.1 Basic Key Area

Esc	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12		
401	458	460	461	462	463	464	465	466	467	468	469	470		
`	1	2	3	4	5	6	7	8	9	0	-	=	\	Back
441	402	403	404	405	406	407	408	409	410	411	412	413	443	414

Tab 415	Q 416	W 417	E 418	R 419	T 420	Y 421	U 422	I 423	O 424	P 425	[426	] 427	Enter 428
Cap s 458	A 430	S 431	D 432	F 433	G 434	H 435	J 436	K 437	L 438	; 439	' 440		
Shif t	Z 444	X 445	C 446	V 447	B 448	N 449	M 450	, 451	. 452	/ 453	Shift 454		
Ctrl 429		Alt 456		Space 457							Alt 456		Ctrl 429

1.3.2 Extension Key Area

Print 470	Scroll 470	Pause		Num 469	/ 453	* 455	– 474
Insert 482	Home 471	Up 473		7 Home 471	8↑ 472	9 Up 473	+ 478
Delete 483	End 479	Down 481		4 ← 475	5 476	6 → 477	
	↑ 472			1 End 479	2↓ 480	3 Down 481	Enter 428
← 475	↓ 480	→ 477		0 482		. 483	

1.4 R Register Interface

1.4.1 Resources Distribution Table

Number	Instructions	Reading/ Writing Regulations (HMI、PLC、 MACRO)	Bit Stora ge	Permane nt Preservat ion
R0~R39	CNCSystem Interface	Defined according to the charts below	Yes	No
R40~R49	PLC Alarm Message			
R50~R80	User-defined	Read/Write	Yes	No
R81~R100	Correspond to system parameters 3401~3420	Read Only	Yes	Yes
R101~R102	Tool Status (save in FRAM)	Read/Write	Yes	Yes
R103~R192	Correspond to milling machine tool data T1~T90 (saved in FRAM) User-defined (non milling machine)	Read/Write	Yes	Yes
R193~R255	User-defined (save in FRAM)			
R256~R511	User-defined			No
R512~R639	CNCSystem Interface	Defined according to the charts below	Yes	No
R640~R1023			No	
R1000	Part Count	Read/Write	No	Yes
R1002	Required Part Count	Read/Write		

Number	Instructions	Reading/ Writing Regulations (HMI、PLC、 MACRO)	Bit Storage	Permanent Preservation
R1004	Total Part Count	Read/Write		
R1005~R1007	Controller ASCII Code	Read Only		
R1009~R1019	CNC system machining time/up time, please refer to the descriptions of each R value.	Read Only		
R1024~R1032	Milling Machine: Occupied by the milling machine auto middle function/auto tool presetting and measuring function. Lathe: User-Defined	Read/Write	No	No
R1033~R2049	User-defined	Read/Write	No	No
R2050~R2090	Pr3810=0: User-defined	Read/Write	No	No
	Pr3810=1: Occupied by the concurrent multiple M codes executing function			
R2091~R2099	User-defined	Read/Write	No	No
R2100~R3100	Pr3228=0: User-defined	Read/Write	No	No
	Pr3228=1: Occupied by the tool managing function			Yes
R3101~R4095	User-defined	Read/Write	No	No
R4096~R5999	CNC System Interface	Defined according to the charts below	No	No

Number	Instructions	Reading/ Writing Regulations (HMI、PLC、 MACRO)	Bit Storage	Permanent Preservation
R6000~R7999	User-defined (save in FRAM)	Read/Write	No	Yes
R8000~R8500	Correspond to system parameters 5001~5500	Read Only	No	No
R8501~R8999	CNC System Interface	Read Only	No	No
R9000~R9999	Robot Kernel Operating Area (Not Customized Area)	Read/Write	No	No
R10000~R10999	CNC system value adding applications memory area	Read/Write	No	Yes
R11000~R14999	CNC System Interface	Read Only	No	No
R15000~R19909	CNC system value adding applications non-memory area	Read/Write	No	No
R20000~R65535	User-defined	Read/Write	No	No

NOTE:

1. The permanent storage of R6000~R7999 is in Fram. If using PIO3, simulator, or controller using PIO5 and Pr5's setting is 9 (Virtual I/O Card), system will not be able to make permanent preservation.
2. DOS system only supports R0~R7999.
3. The part of this table about the macro program is only suitable for read and write R values using G10 L1000 P_R_; if you use the @ method to read and write R values in the macro program, please refer to the [Macro 变数规格](#)¹.

1.4.2 Interface Instructions

No	Title & Range	Function Description	Read/ Write Principle
----	---------------	----------------------	-----------------------------

¹ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=61408070>

R0	Marking special edition (ISSUE-ID + 2 version codes)	Records the special edition version of the controller installation package.	Read only
R1	M Code (00~99)	When the CNC executes M CODE, it stores the contents of the M CODE here for the LADDER to read.	Read only
R2	S Code (0000~9999)	When the CNC executes S CODE, it stores the contents of the S CODE here for the LADDER to read.	Read only
R3	T Code (0000~9999)	When the CNC executes T CODE, it stores the contents of the T CODE here for the LADDER to read.	Read only
R4		Synthesis feedrate command value. Unit: LIU/min	
R5	B Code (0000~9999)	The function can be enabled by Pr 3806 When CNC executes B CODE, it stores the contents of the B CODE here for the LADDER to read. Only the 1 st MST channel supports the execution of B CODE on PLC. (Refer to Pr721~Pr730)	Read only
R6~R12	Reserved for CNC	System Reserved Area	

R13®	Mode Selection	Machining mode selection 1: Edit 2: Auto 3: MDI 4: JOG 5: INCJOG 6: MPG 7: HOME	Write only
R14	Increment Feed	Incremental JOG mode and MPG mode segments settings, this R value will not be affected by Pr3207. INCJOG Mode: 1: x1 LIU 2: x10 LIU 3: x100 LIU 4: x1000 LIU MPG Mode: 1: x1 LIU 2: x10 LIU 3: x100 LIU 4: The magnified rate is set in Pr2001 Note: Please refer to Pr17² for the definition of LIU.	Write only

² https://confluence.syntecclub.com.tw/display/TechManual/Pr17+*Control+precision

R15	Spindle Override	Spindle segment setting (Maximum value is 200%) 1. If Pr3207=1, R15 is the percentage of spindle override, i.e. R15=5 means spindle override is 5%. 2. If Pr3207=2, override segment defined as below: R15=1, spindle override is 50% R15=2, spindle override is 60% R15=8, spindle override is 120% 3. The actual spindle override is $R15 \times R851\sim$, which the unit of R851~ is defined as percentage, it does not affected by Pr3207~	Write only
R16	Feedrate Override	G01,G02,G03 segment setup 1. If Pr3207=1, R16 is the percentage of spindle override, e.g. R16=5 means spindle override is 5%. 2. If Pr3207=2, R16 definition as below: R16=1, spindle override is 10% R16=2, spindle override is 20% R16=20, spindle override is 200%	Write only
R17	JOG Override	JOG segment setup 1. If Pr3207=1, R17 is the percentage of jog override, e.g. R17=5 means spindle override is 5%. 2. If Pr3207=2, R17 definition as below: R17=1, spindle override is 10%. R17=2, spindle override is 20%, R17=20, spindle override is 200%. 3. The lower limit of R17 is 0, JOG Override will be 0% if R17 is set to be a negative value; there is no upper limit for R17, but the speed of axis is limited by Pr461~	Write only

R18	Rapid Traverse Override	G00 segment setup 1. If Pr3207=1, R18 is the percentage of spindle override, e.g. R18=1 means spindle override is 1%. 2. If Pr3207=2, R18 definition as below: R18=1, spindle override is F0. R18=2, spindle override is 25%. R18=3, spindle override is 50%. R18=4, spindle override is 100%.	Write only
R19	Chopping Feedrate Override	Chopping federate segments settings 1. If Pr3207=1, the feeding percentage equals to the content of R19. e.g. R19=5 means the feedrate override is 5%. 2. If Pr3207=2, the R19 definition as below: R19=1 represents 10% R19=2 represents 20% R19=15 represents 150%	Write only
R20	Servo Axis Enabled	Servo Axis Enabled Bit 0 reserved, not using. Bit1 represents the 1stAxis (X) enabled. Bit 2 represents the 2ndAxis (Y) enabled. Bit18 represents the 18thAxis enabled.	Read only

R21	Active Session ID	Multi-Path CNC main system executing mode: 0: Execute the Multi-Path program or run the general non-Multi-Path program in the 1st path. 1: Only execute the \$1 program or general non-Multi-Path program in the 1st path. 2: Only execute the \$2 program or general non-Multi-Path program in the 2nd path. 3: Only execute the \$3 program or general non-Multi-Path program in the 3rd path. 4: Only execute the \$4 program or general non-Multi-Path program in the 4th path. Function expansions from 10.116.34, 10.117.34. If R21.11 is OFF, the specifications remain the same as above; if R21.11 is ON, change the specifications as below: Bit 0, 12-31 reserved, not using Bit 1: execute the \$1 program or general non-Multi-Path program in the 1st path. Bit 2: execute the \$2 program in the 2nd path. Bit 3: execute the \$3 program in the 3rd path. Bit 4: execute the \$4 program in the 4th path. Bit 5-10: not using, only reserved for the possibility of future expansion to 10 axis. Bit 11: R21 definition switch. R21 expansion function (R21.11 on) remarks; • R21.1~R21.4 OFF: All the paths won't CycleStart when activated C0, suggest the developing personnel to provide proper protection for the action in PLC. • Pr731=3 (The 1/2/3 path is the main system path), if R21.1, 2, R21.4 are ON, only the 1st and 2nd paths CycleStart when activated C0.	Read only
------------	-------------------	--	-----------

		Switching R21 during machining/pause/block stop does not effect to the actual machining paths and HMI display, suggest the developing personnel to provide proper protection for the action in PLC.	
R22	MPG: 1: x1 2: x10 3: x100 4: Base on Pr2001.	The 2nd MPG segment selection	Write only
R23	MPG: 1: x1 2: x10 3: x100 4: Base on Pr2001.	The 3 rd MPG segment selection	Write only
R24	Spindle Gear Selection	Spindle gear shifting settings: 1 represents the 1st gear 2 represents the 2nd gear 3 represents the 3rd gear 9 represents the blank gear.	Write only
R25	Actual feedrate		
R26	Program Coordinate X	X axis program coordinate position, the unit is the minimum input unit, LIU. When control precision is micro meter, its value would be: 0.01 mm 0.01 deg, 0.0001 inch	Read only
R27	Program Coordinate Y	(As above)	Read only

R28	Program Coordinate Z	(As above)	Read only
R29	Program Coordinate C(6 th)	(As above)	Read only
R30	IO Signal of 1 ^s t Driver Set	R30.1~R30.18 correspond to the 1stAxis~ 18thAxis driver IO signal. Note 1: Supporting 10.116.20, 10.117.21 or above version, that is supporting the torque arrival signal. Note 2: Torque arrival signal is only supported by; 1. YASKAWA M2/M3 2. SYNTEC M2/M3 3. EtherCAT, for brands and version info refer to EtherCAT Driver Application Manual³ Note 3: To activate the function, set the driver parameters first to enable the torque limitation function and restart the driver. Note 4: R30 was preset to monitor the torque arrival signal, via Pr3266~Pr3267 Select the drive signal to be monitored⁴ (External 1 signal or Brake signal) Pr3266 = 1, R30.1~R30.18 corresponds to the torque arrival signal of the 1st~16thaxis, the Bit becomes 1 when the torque arrives the limit value. Pr3266 = 2, R30.1~R30.18 corresponds to the EXT1 signal of the 1st~16thaxis, the Bit becomes 1 when the driver receives the EXT1 input signal. Pr3266 = 3, R30.1~R30.18 corresponds to the Brake signal of the 1st~16thaxis, the Bit becomes 1 when the driver activates the brake. Supporting 10.116.36Y, 10.116.51, 10.117.51 and above version. Note 5: Only YASKAWA M2/M3 supports the External 1 signal and the Brake signal.	Read only

³ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pagelId=379437501>

⁴ <https://confluence.syntecclub.com.tw/display/TechManual/Pr3266~Pr3267+Select+the+drive+signal+to+be+monitored>

R31	Machine Coordinate X	X axis machine coordinate position, the unit is the minimum length input unit: LIU, same as the length unit of system parameters.	Read only
R32	Machine Coordinate Y	(As above)	Read only
R33	Machine Coordinate Z	(As above)	Read only
R34	Machine Coordinate C(6 th)	(As above)	Read only
R35	Spindle Machine Position	Spindle machine position. Unit: 0.001 degree.	Read only
R36	1 st Spindle Speed	Current 1 st spindle speed according to the feedback value or the command value if there is no encoder. If the 1 st spindle is not activated, the value is fixed to 0 RPM. Unit: RPM.	Read only
R37	Movement direction relative to current working plane	Movement direction of tool relative to current working plan, unit is 0.001 degree.	Read only
R38	Argument one	Argument one, the meaning is defined by the C Bit function that uses this argument.	Write Only
R39	Relay 2 A/D value	Relay2 A/D converter value	Read only
R40	PLC Alarm Message #1 ~ #16 #251 ~ #266	System default alarm area, corresponds to alarm number 1 to 16, please refer to the mechanical logic control alarm list for definitions. Bit 0~15 #1 ~ #16 PLC Alarm Message, save to history. Bit16~31 #251 ~ #266 PLC Warning Message, no save to history.	Write Only

R41	PLC Alarm Message #17 ~ #32 #267 ~ #282	System default alarm area, corresponds to alarm number 17 to 32, please refer to the mechanical logic control alarm list for definitions. Bit 0~15 #17 ~ #32 PLC Alarm Message, save to history. Bit16~31 #267 ~ #282 PLC Warning Message, no save to history.	Write Only
R42	PLC Alarm Message #33 ~ #48 #283 ~ #298	System default alarm area, corresponds to alarm number 33 to 48, please refer to the mechanical logic control alarm list for definitions. Bit 0~15 #33~ #48 PLC Alarm Message, save to history. Bit16~31 #283 ~ #298 PLC Warning Message, no save to history.	Write Only
R43	PLC Alarm Message #49 ~ #64 #299 ~ #314	System default alarm area, corresponds to alarm number 49 to 64, please refer to the mechanical logic control alarm list for definitions. Bit 0~15 #49 ~ #64 PLC Alarm Message, save to history. Bit16~31 #299 ~ #314 PLC Warning Message, no save to history.	Write Only
R44	PLC Alarm Message #65 ~ #80 #315 ~ #330	User define alarm message, corresponds to alarm number 65 to 80. Bit 0~15 #65 ~ #80 PLC Alarm Message, save to history. Bit16~31 #315 ~ #330 PLC Waring Message, no save to history.	Write Only
R45	PLC Alarm Message #81 ~ #96 #331 ~ #346	User define alarm message, corresponds to alarm number 81 to 96. Bit 0~15 #81 ~ #96 PLC Alarm Message, save to history. Bit16~31 #331 ~ #346 PLC Warning Message, no save to history.	Write Only
R46	PLC Alarm Message #97 ~ #112 #347 ~ #362	User define alarm message, corresponds to alarm number 97 to 112. Bit 0~15 #97 ~ #112 PLC Alarm Message, save to history. Bit16~31 #347 ~ #362 PLC Warning Message, no save to history.	Write Only
R47	PLC Alarm Message #113 ~ #128 #363 ~ #378	User define alarm message, corresponds to alarm number 113 to 128. Bit 0~15 #113 ~ #128 PLC Alarm Message, save to history. Bit16~31 #363 ~ #378 PLC Warning Message, no save to history.	Write Only

R48	PLC Alarm Message #129 ~ #144 #379 ~ #394	User define alarm message, corresponds to alarm number 129 to 144. Bit 0~15 #129 ~ #144 PLC Alarm Message, save to history. Bit16~31 #379 ~ #394 PLC Warning Message, no save to history.	Write Only
R49	PLC Alarm Message #145 ~ #160 #395 ~ #410	User define alarm message, corresponds to alarm number 145 to 160. Bit 0~15 #145 ~ #160 PLC Alarm Message, save to history. Bit16~31 #395 ~ #410 PLC Warning Message, no save to history.	Write Only
R50~R80	User define	User-defined	Read/Write
R81~R100	MLC System Parameter	Apply MLC system parameter No.3401~No3420 to build a LADDER which is adaptable to various machines, such as tool changing and etc.	Read Only
R101	Tool Magazine Position	Tool Magazine position status	Read/Write
R102	Current Spindle Tool Number	Current Spindle Tool Number Used in the HMI display of T code in single program system (Pr731=1); please refer to R525 for multi-program system.	Read/Write
R103~R192	Mill Tool Data	Correspond to milling machine tool data T1~T90 (saved in FRAM) User-defined (non milling machine)	Read/Write
R193~R255	User define	User-define (saved in FRAM)	Read/Write
R256~R511	User define	User-define	Read/Write

R512	Net Ready	Net PLC Function, when the corresponding bit is ON, means the station in network is READY. All the shared data are the latest. Bit 0: Reserved. Bit 1: Station 1 ready Bit 2: Station 2 ready Please check Pr3236~Pr3238 for Net PLC setups.	Read Only
R513	Net Timeout	Net PLC Function. The network should have been timeout when the corresponding Bit shows ON. It's an exception if just started up the system, but if not receiving any updates of the station within 100ms after the connection, then it'll be a timeout. The information update frequency is decided by the PLC scanning time. Bit 0: Reserved. Bit 1: Station 1 timeout. Bit 2: Station 2 timeout. 	Read Only
R514	Pending alarm module ID	Pending alarm module ID of current PLC display. Updates the pending alarm module information to PLC every 3 seconds by HMI. 1: PLC 2: Coordinate 3: Motor 4: Spindle 5: Compiler 6: Operation 7: PLC Hint 8: Macro There is no alarm module code for serial servo drives, if needed, please refer to S171~186.	
R515	Pending alarm ID	Pending alarm ID of current PLC display. Updates the pending alarm information to PLC every 3 seconds by HMI.	

R516	Pending alarm Axis ID	Pending alarm Axis ID of current PLC display. Updates the pending alarm information to PLC every 3 seconds by HMI.	
R517	Pending alarm Spindle ID	Pending alarm Spindle ID of current PLC display. Updates the pending alarm information to PLC every 3 seconds by HMI.	
R518	Manual Coordinate (5 Axis RTCP)	Define the coordinate system referring to when giving manual commands (MPG、JOG、INCJOG) 0: Machine Coordinate 1: Program Coordinate (G92 Coordinate) 2: Tool Coordinate (Only effective in five-axis machine.)	
R519	Manual RTCP	Manual RTCP Set R519 to 1 and execute G43.4 to enable RTCP.	

R520~ R529	First Axis Group Control Registers	First path command registers. When the path belongs to the CNC main system, the register commands with turn to drive the CNC main system. Registers: R520 Command Register R521 State Register R522 Program No. (0~999999) R523 Feedrate Override (-200%~200%) R524 Rapid travel Override (-100~100) R525 Target Tool Number. Used in the HMI display of the first main system path T code in multi-program system (Pr731#1). Please refer to R102 for single program system. R526 Remaining block distance. R527~529 Reserved for future expansion. Program Number Guideline: • R522 correspond to 1~999999 and to O0001~O999999. • When the path belongs to the CNC main system. 1. When R522 is 0, the file name will be decided by HMI. Hence R520.0 will be the same as C0 action. 2. When R522 isn't 0, the file name will be decided by HMI, Hence R520.0 will Cycle Start the program designated by R522. 3. The program designated by R522 allows to use \$1, \$2, which are multi-path programs. 4. The CNC main system can only be activated by the first path. None of the other main system paths can start the system. • When the path does not belong to the CNC main system, which means it's only a simple PLC path, it will do nothing if the value is set to be 0, and the file name will be decided by R522 if the value isn't 0. Feedrate Guideline: The register won't work if the path belongs to the CNC main system. The main system takes R16 and R18 as its feedrate reference. Command Register Guideline (the bit out of below description No. is reserved):	
-----------------------	--	---	--

No.	Command	Description	Attention note
R520.0	Cycle Start	The axis group execute specified program.	Please refer to the program numbers above for the designation of executing program.
R520.1	Feedhold	Feedhold the axis group movement.	
R520.2	Reset/ Abort	- Pr741 is 0 - Only reset the internal status of axis group. - Pr741 is 1 - if it is the axis group of independent operation, then reset for axis and spindle under axis group, and clear alarm. - Otherwise, only reset the internal status of axis group.	- Please refer to Pr741 *The rule of non-primary axis group⁵ for setting of axis group of independent operation. Pr741 can be set to 1, only if the system is 10.118.40 and later version.

⁵ https://confluence.syntecclub.com.tw/display/TechManual/Pr741+*The+rule+of+non-primary+axis+group

No.	Command	Description	Attention note
R520.3	MPG Simulation		
R520.4	Block Optional Skip		Block optional skip is not compatible with all G code format, a COM alarm will appear when the unsupported format detected.
R520.5	Emergency Stop	- Pr741 is 0 - Not supported. - Pr741 is 1 - The bit is supported only for axis group of independent operation.	- Please refer to Pr741 *The rule of non-primary axis group⁶ for setting of axis group of independent operation. Pr741 can be set to 1, only if the system is 10.118.40 and later version.

⁶ https://confluence.syntecclub.com.tw/display/TechManual/Pr741+*The+rule+of+non-primary+axis+group

No.	Command	Description	Attention note
		When C36 is on, the axis group state will be not ready. And when C36 is off, the axis group use this bit to further select ready/not ready state.	
R520.11 ~R520.19	Block Optional Skip #, '/#'	- R520.11 Block Optional Skip 1, '/1' - R520.12 Block Optional Skip 2, '/2' - R520.13 Block Optional Skip 3, '/3' - R520.14 Block Optional Skip 4, '/4' - R520.15 Block Optional Skip 5, '/5' - R520.16 Block Optional Skip 6, '/6' - R520.17 Block Optional Skip 7, '/7' - R520.18 Block Optional Skip 8, '/8'	- Only support 10.114.20 and later version. - Block optional skip is not compatible with all G code format, a COM alarm will appear when the unsupported format detected.

No.	Command	Description	Attention note
		R520.19 Block Optional Skip 9, '/9'	

State Registers guideline (the bit out of below description No. is reserved):

No.	State	Description	Attention note
R521.0	Busy	When the system is busy, this bit will be on.	
R521.1	Feedhold	When the system is feedhold, this bit will be on.	
R521.2	Work	When a system is busy(S0, R521.0 on), feedhold(S1, R521.1 on) or block stop(S2 on), this bit will be on.	
R521.3	NC Ready	- Pr741 is 0 - When there are "C36 on" or "system in alarm", this bit will be off.	- Only support 10.118.40 and later version. - Please refer to Pr741 *The rule of non-primary axis group⁷ for setting of axis group of independent operation. - Pr741 can be set to 1, only if the system is 10.118.40 and later version.

⁷ https://confluence.syntecclub.com.tw/display/TechManual/Pr741+*The+rule+of+non-primary+axis+group

No.	State	Description	Attention note
		- When there are "C36 off" and "system no alarm", this bit will be represented for the all axis under axis group whether are all servo on. - Pr741 is 1 - Only for axis group of independent operation to extend rule, and others axis group remain original rule the same as Pr741 is set to 0.	

No.	State	Description	Attention note
		When the re are "C3 6 on", "In dependent axis group in alarm" or "R 520.5 on", this bit will be off.	

No.	State	Description	Attention note
		When the re are "C3 6 off", "In dependent axis group no alarm" and "R52 0.5 off", this bit will be represented for the all axis under axis group whether are all servon.	

No.	State	Description	Attention note
R521.4	Servo Not Ready	It represents that whether there is any servo not ready under the axis group.	Only support 10.118.40 and later version.
R521.5	System Alarm	- Pr741 is 0 - Always be 0 (not support). - Pr741 is 1 - Only support for axis group of independent operation, and others axis group, non independent operation, are always 0.	- Only support 10.118.40 and later version. - Please refer to Pr741 *The rule of non-primary axis group⁸ for setting of axis group of independent operation. - Pr741 can be set to 1, only if the system is 10.118.40 and later version.

⁸ https://confluence.syntecclub.com.tw/display/TechManual/Pr741+*The+rule+of+non-primary+axis+group

No.	State	Description	Attention note
		This bit represents that whether axis group of independent operation has "system type" or "axis group type, axis type (including servo driver type)" alarm in happening.	
R521.6	System Warning	- Pr741 is 0 - Always be 0 (not support). - Pr741 is 1 - Only support for axis group of independent operation, and others axis group, non independent operation, are always 0.	

		No.	State	Description	Attention note
				This bit represents that whether axis group of independent operation has "system type" or "axis group type, axis type (including servo driver type)" warning in happening.	
		R521.7	Block stop	During single block stop(BLOCK STOP status), this bit will be on.	Only support 10.118.29W, 10.118.40C, 10.118.42 and later version.
		Remaining block distance : Show the remaining block distance while the path is executing, this value is read only and will always bigger than 0. Unit: BLU			
R530~R539	Second Axis Group Control Registers	Second path command registers. The register commands won't work when the path belongs to the CNC main system. R530 Command Register R531 State Register R532 program No. (0~999999) R533 Cutting Feedrate Override (-200%~200%) R534 Rapid travel Feedrate Override (-100~100) R535 Target Tool Number R536 Remaining block distance R537~539 Reserved for future expansion			

R540~ R549	Third Axis Group Control Registers	Third path command registers. The register commands won't work when the path belongs to the CNC main system. R540 Command Register R541 State Register R542 program No. (0~999999) R543 Feedrate Override (-200%~200%) R544 Rapid travel Override (-100~100) R545 Target Tool Number R546 Remaining block distance R547~549 Reserved for future expansion	
R550~ R559	Fourth Axis Group Control Registers	Fourth path command registers. The register commands won't work when the path belongs to the CNC main system. R550 Command Register R551 State Register R552 program No. (0~999999) R553 Feedrate Override (-200%~200%) R554 Rapid travel Override (-100~100) R555 Target Tool Number R556 Remaining block distance R557~559 Reserved for future expansion	
R560~ R579	Reserved for CNC	System reserved area	
R580	Spindle PLC Control Mode	Spindle PLC Control Mode Under this mode, the spindle command will be controlled by PLC and decided by R941~R950 (Correspond to the 1st~10th spindle). It represents the PLC control mode when the corresponding Bit is 1; general CNC control mode when the corresponding Bit is 0. Bit 0, 11-31 → Reserved Bit 1 → 1st spindle controlled by PLC Bit 2 → 2nd spindle controlled by PLC Bit 10 → 10th spindle controlled by PLC	

R581	Spindle OFF	Spindle Mode Command The corresponding Bit (R581.x) only works when the parameter 179x sets up the spindle motor type to servo motor. When the corresponding bit is 1, the logic axis switches to position orientation control mode; when the corresponding bit is 0, the logic axis switches back to spindle control mode, the clockwise/counterclockwise rotation control is defined by R583/R584. Bit 0, 1, 11-31 → Reserved Bit 2 → 2nd spindle mode command. Bit 3 → 3rd spindle mode command. ... Bit 10 → 10th spindle mode command. Note: - 1st spindle mode command is controlled by C63. - When using M code to operate the Spindle CS axis exchange, the action is directly done by the kernel (not with C38).	Write Only
R582	Spindle axis mode	Spindle current axis mode When the corresponding bit is 1, the spindle is under position orientation control mode; when the corresponding bit is 0, the spindle is under spindle speed control mode. Bit 0, 11-31 → Reserved Bit 1 → 1st spindle axis control mode Bit 2 → 2nd spindle axis control mode ... Bit 10 → 10th spindle axis control mode	Read Only

R583	Rotating the spindle clockwise	Spindle clockwise rotation Bit 0, 1, 11-31 → Reserved Bit 2 → 2nd spindle clockwise rotation. Bit 3 → 3rd spindle clockwise rotation. ... Bit 10 → 10th spindle clockwise rotation. Note: 1. 1st spindle clockwise rotation is controlled by C64. 2. To use the spindle rotating speed arrival check function correctly (parameter 1991~), it is necessary to activate the R bit to switch the system spindle into operating mode even it is a VFD spindle.	Write Only
R584	Rotating the spindle counterclockwise	Spindle counterclockwise rotation Bit0,1, 11-31 → Reserved Bit 2 → 2nd spindle counterclockwise rotation. Bit 3 → 3rd spindle counterclockwise rotation. ... Bit 10 → 10th spindle counterclockwise rotation. Note: 1. 1st spindle counterclockwise is controlled by C65. 2. To check spindle speed arrival, this bit must be triggered even if it is a inverter spindle. 3. To use the spindle rotating speed arrival check function correctly (parameter 1991~), it is necessary to activate the R bit to switch the system spindle into operating mode even it is a VFD spindle.	Write Only
R585	Spindle JOG Mode	Spindle JOG Mode Command When the corresponding bit is 1, it forces the spindle to rotate with the spindle JOG speed set by parameter 172x. The function is usually used on spindle tool change and modeling of Milling machine or auto feed of Lathe. Bit 0, 1, 11-31 → Reserved Bit 2 → 2nd spindle JOG mode Bit 3 → 3rd spindle JOG mode ... Bit 10 → 10th spindle JOG mode. Note: 1st spindle JOG mode is controlled by C60.	Write Only

R586	Spindle Orientation Stop	Spindle Orientation Mode Command When the corresponding bit is 1, it runs the spindle orientation function. The function only works with an encoder installed on the spindle and the corresponding bit remains ON. Parameter 174x: Modify the spindle orientation speed. Parameter 175x: Modify the spindle orientation angle, the angle is the synchronization signal position relative to the encoder. Therefore, the parameter must be readjusted every time after changing a motor, encoder or belt. Parameter 176x: Modify the spindle orientation time. R587: Show the complete signal of the spindle orientation function. Bit 0, 1, 11-31 → Reserved Bit 2 → 2nd spindle orientation mode. Bit 3 → 3rd spindle orientation mode. ... Bit 10 → 10th spindle orientation mode. Note: 1st spindle Orientation is controlled by C61	Write Only
R587	Spindle Orientation Stop Finish	Spindle Orientation finish flag Bit 0, 1, 11-31 → Reserved Bit 1 → 1st spindle orientation finish Bit 2 → 2nd spindle orientation finish Bit 3 → 3rd spindle orientation finish ... Bit 10 → 10th spindle orientation finish Note: S61 & R587.1 both are flags of 1st spindle.	Read Only
R588	Rigid Tapping Mode	Spindle rigid tapping mode Bit 1 → 1st spindle is in rigid tapping mode Bit 2 → 2nd spindle is in rigid tapping mode Bit 3 → 3rd spindle is in rigid tapping mode ... Bit 10 → 10th spindle is in rigid tapping mode	

R589	Multiple Axis Group MST associated information	This Registry stores M/S/T information triggered by S29/S54/S69, and the information will be contained until next M/S/T command is read. [Bit 1 ~ 4, M/S/T Command Path Information] R589.1 The current M/S/T is being sent by \$1 (1st path). R589.2 The current M/S/T is being sent by \$2 (2nd path). ... [Bit 11 ~ 20 Spindle Information] R589.11, The current M/S/T is being sent by \$x, its main spindle is the first spindle. R589.12, The current M/S/T is being sent by \$x, its main spindle is the second spindle. ... R589.20, The current M/S/T is being sent by \$x, its main spindle is the tenth spindle.	
R590~ R599	PLC warning message	PLC Hint(Warning) Message Area, each bit triggers one hint. There are 32 bits in one register, and a total 320 messages can be used. R590.0 Reserved R590.1 for 1st hint message R590.2 for 2nd hint message .. R590.31 for 31st hint message R591.0 for 32nd hint message .. R592.0 for 64th hint message .. R593.0 for 96th hint message .. Hint message is defined from string list 23001. That is, 23001 is the content of 1st hint, 23002 is the content of 2nd hint, so on and so forth.	

R600	Machine Lock	Machine Lock Bit 0 Reserved Bit 1 for 1st axis (X) lock Bit 2 for 2nd axis (Y) lock ... Bit 18 for 18th axis lock ... - The axis won't be moving if the command (e.g. G91G00X100) is sent under this function being activated (R600.X ON), both the remaining distance and the program coordinate will show normally, but the machine coordinate will be fixed till the function being turned off, then the system will synchronize the machine coordinate with the program coordinate. - If this function being activated (R600.X.ON) during axial movements, the axis won't be moving and both the remaining distance and the program coordinate will show normally, but the machine coordinate will be fixed right after the function being activated till the function being turned off, then the system will synchronize the machine coordinate with the program coordinate.	Write Only
R601	Axis MPG simulation	Under the PLC Axis program, if the corresponding flag bit is 1, the feedrate speed of the PLC axis is decided by the rotating speed of MPG. The machine moves faster when the MPG rotates faster; the machine moves backwards when the MPG is rotating reversely; the machine stops when the MPG stops. It is suitable for trial machining. Bit 0 Reserved. Bit 1 for 1st axis (X) MPG simulation Bit 2 for 2nd axis (Y) MPG simulation ... Bit 18 for 18th axis MPG simulation	
R602	Spindle Y-Delta Request Mode	Spindle Y-Delta request mode, the switching speed is set by parameter 1841~1850. If the corresponding bit is 0 means Y connect; 1 means Delta connect. Bit 0 Reserved. Bit 1 is 1st axis Y-Delta connect Mode. Bit 2 is 2nd axis Y-Delta connect Mode. ... Bit 10 is 10th axis Y-Delta connect Mode.	

R603	Inhibit Axis Command	Inhibit specific axis moving command. When the Inhibit Axis command is ON, the running program turns into Feedhold state. R841~R844 will show the feedhold reason. (Versions after 10.116.38B, 10.116.52E, 10.116.54, 10.117.54 turn into Feedhold state, previous versions turn into Block Stop state) Bit 0: Reserved. Bit 1: Inhibit 1st Axis Motion Command. Bit 2: Inhibit 2nd Axis Motion Command. ... Bit 18: Inhibit 18th Axis Motion Command.	
R604	Teach In command	Teach In Command Area Bit 0 Rapid orientation endpoint teaching Bit 1 Straight cutting endpoint teaching Bit 2 Arc cutting teaching. First trigger is the arc middle point input, second trigger is the arc endpoint input. Bit 3 Point coordinate teaching. Bit 7 Designate the arc teaching plane to be G17, and add G17 in front of G codes when generating. Bit 8 Designate the arc teaching plane to be G18, and add G18 in front of G codes when generating. Bit 9 Designate the arc teaching plane to be G19 and add G19 in front of G codes when generating. Bit 15 Edit Overwrite Mode. On: Overwrite the line where cursor at; Off: insert line. <Note> R604.x Teach In Command is only allowed to be used in F2 Program Edit -> F6 Teaching Page	

R605	Tool Offset Write	Tool Length Measurement Write In Command Bit 0 Tool Length Measuring Function activated. Bit 1: Z+ contacting surface write in request. Bit 2: Z- contacting surface write in request. Bit 3: X+ contacting surface write in request. Bit 4: X- contacting surface write in request. ...	
R606	MPG Shift	MPG Shift Enable Command 1. R606=0 Disable MPG Shift R606=1 Enable MPG Shift 2. Caution: When MPPG shift enabled, with C16~ to assign MPG axis, the axis can be moved by MPG under any circumstances. Please be extra careful!	Write Only
R607	Spindle Speed Arrival	Spindle Speed Arrival Signal, identical to S64/S59. Bits 0, 1, 11, 21-31 Reserved. 1. Under cutting G01/G02/G03 R607.2 means 2nd Spindle speed arrival R607.3 means 3rd Spindle speed arrival ... R607.10 means 10th Spindle speed arrival 2. Under threading R607.12 means 2nd Spindle speed arrival R607.13 means 3rd Spindle speed arrival ... R607.20 means 10th Spindle speed arrival	Read Only

R608	Modbus PLC Communication	R608.0 Trigger this Bit to show that the master is ready for Modbus communication. (R608.10, R608.31, and error code R5039 will be cleared when the PLC Modbus component is triggered .) R608.1 (Read Only) Transmitting Status. R608.2 (Read Only) Waiting to Receive. R608.3 (Read Only) Receiving. R608.10 (Read Only) Reception finished correctly. R608.31 (Read Only) Error Flag . ON indicates that error(s) occurred and the error code will be displayed at R5039.	
R609	Axis Coupling Status	Axis Coupling Status When coupling type (Pr3825/Pr3865/Pr3875/Pr3885) ≠1 (non-mechanical coupling type), the Register displays coupling status. If the coupling status is changed by C46 or R614, this Register will update after system finishes mode switch. Bit 1: 1st set of axis is coupling. Bit 2: 2nd set of axis is coupling. Bit 3: 3rd set of axis is coupling. Bit 4: 4th set of axis is coupling. Note: Please refer to C46/R614 Axis Coupling Switch for axis coupling control.	
R610	Skip Position Latched	G31 or G28.1 skip command touch the skip signal and record the position. Bit 0: Reserved Bit 1: 1st axis (X) skip occur Bit 2: 2nd axis (Y) skip occur ... Bit 18: 18th axis skip occur	Read Only
R611	Axis Movement Flag	Display each axis movement status, according to the feedback value, failing the Zero Speed Check means the axis is moving. Zero Speed Check parameters are Pr901-920. Bit 0 Reserved Bit 1 : 1st axis (X) is moving Bit 2 : 2nd axis (Y) is moving ... Bit 18 : 18th axis is moving	

R612	Machine Positive Direction Lock	Machine Positive Direction Lock. Bit 0 Reserved Bit 1 : 1st axis (X) positive direction lock. Bit 2 : 2nd axis (Y) positive direction lock. ... Bit 18 : 18th axis positive direction lock. • The axis won't be moving if the command (e.g. G91G00X100) is sent under the function being activated (R612.xx ON), and both the program and the machine coordinate will be fixed. • If the function being activated (R612.x ON) during axial movements, the axis will stop moving immediately, and both the program and the machine coordinate will be fixed right after the function being activated.	
R613	Machine Negative Direction Lock	Machine Negative Direction Lock. Bit 0 Reserved Bit 1 : 1st axis (X) negative direction lock. Bit 2 : 2nd axis (Y) negative direction lock. ... Bit 18 : 18th axis negative direction lock. • The axis won't be moving if the command (e.g. G91G00X100) is sent under the function being activated (R612.xx ON), and both the program and the machine coordinate will be fixed. • If the function being activated (R612.x ON) during axial movements, the axis will stop moving immediately, and both the program and the machine coordinate will be fixed right after the function being activated.	
R614	Axis Coupling Request	Axis coupling status switch. If Coupling Type (Pr3865/Pr3875/Pr3885) ≠1 (non-mechanical coupling), the actual coupling mode is decided by this Register. When FLAG ON, the coupling axis direction turns into coupling status; when FLAG OFF, the coupling axis direction leaves coupling status. Under coupling status, the slave axis moves with the set ratio when the spindle moves. Bit 0, 1 Reserved Bit 2 → 2nd set of axis coupling command Bit 3 → 3rd set of axis coupling command Bit 4 → 4th set of axis coupling command Note: • The 1st set of axis coupling command is activated by C46. • Please refer to R609 Axis Coupling Status for axis coupling status.	

R615~ R618	Second M/S/T Execution Channel	Second M/S/T Execution Channel R615 command and status R616 M code R617 S code R618 T code R615.0 MST code finish (PLC → CNC) R615.1 DEN moving command finish (CNC → PLC) R615.2 M code read (CNC → PLC) R615.3 S code read (CNC → PLC) R615.4 T code read (CNC → PLC) ... R615.8 M/S/T code is executed by 1st path (\$1) R615.9 M/S/T code is executed by 2nd path (\$2) R615.10 M/S/T code is executed by 3rd path (\$3) R615.11 M/S/T code is executed by 4th path (\$4) R615.12 M00 Read Flag R615.13 M01 Read Flag; need to activate the optional stop function (C44). R615.14 M02 Read Flag R615.15 M30 Read Flag Note: 1. Valid version: 116.41, 10.117.41 2. The flag only works when the parameter of “Pr732 * multi-program MST channel rules” is set to 1. 3. The flag will activate automatically after reading the M code till system reset (Reset) or M Code Finish (R615.0 On). 4. To ensure the system activates the flag correctly, R615.0 should be OFF before the flag being activated by the kernel in PLC applications. R615.16 The current M/S/T is being sent by \$x, its main spindle is the 1st spindle. R615.17 The current M/S/T is being sent by \$x, its main spindle is the 2nd spindle. ... R615.23 The current M/S/T is being sent by \$x, its main spindle is the 8th spindle. R615.29...The current M/S/T is being sent by \$x, its main spindle is the 9th spindle. R615.30...The current M/S/T is being sent by \$x, its main spindle is the 10th spindle.	
-----------------------	--------------------------------	--	--

		(When parameter 3810 activates the Single Block multi-M code execution function) R615.24 1st M code execution request R615.25 2nd M code execution request R615.26 3rd M code execution request R615.27 4th M code execution request R615.28 5th M code execution request	
R619~ R622	Third M/S/T Execution Channel	Third M/S/T Execution Channel R619 Control and Status Register R620 M Code R621 S Code R622 T Code	
R623~ R626	Fourth M/S/T Execution Channel	Fourth M/S/T Execution Channel R623 Control and Status Register R624 M Code R625 S Code R626 T Code	

R627	Axis Torque Mode	The axis switch to torque control under the mode, the corresponding commands are R880~R895. When the corresponding bit is 1 means the axis is switched to torque control mode; when the corresponding bit is 0 means the axis is position control mode. R627.0 Reserved, not using. R627.1 represents the 1st axis torque control mode R627.2 represents the 2nd axis torque control mode ... R627.18 represents the 18th axis torque control mode Note 1: The interface only works for serial bus communication. Note 2: The control interface only supports version 10.118.10 and below.	Write Only
R628	Spindle CSS Mode	Display the current spindle cutting speed mode. When the corresponding Bit is ON, the spindle is in the constant surface cutting speed control mode (G96); when the corresponding Bit is OFF, the spindle is in the rotating speed control mode (G97). Bit 0, 5-31 Reserved Bit 1 → machining spindle of the 1st path Bit 2 → machining spindle of the 2nd path Bit 3 → machining spindle of the 3rd path Bit 4 → machining spindle of the 4th path	Read Only

R629	Axis M code	The register provides the PLC axis M code request signal. When axis executes to M codes, the corresponding Bit ON and request new M code from PLC program, M code contents are stored in the R618~R696. After the controller send the request, only until M code completion signal (R630) On, then next block start executing. Bit 0 Reserved Bit 1 → 1st axis (X) M code request Bit 2 → 2nd axis (Y) M code request ... Bit 18 → 18th axis M code request	Read Only
R630		M Code Completion Signal (Please refer to R629) Bit 0 Reserved, not using Bit 1 represents the 1staxis (X) M code completion. Bit 2 represents the 2ndaxis (Y) M code completion. ... Bit 18 represents the 18thaxis M codes completion.	Write Only
R631	The target winding of Y-Delta of each spindle.	Each spindle Y-Delta target state When the corresponding bit is 0, the spindle is Y connection; 1 means Delta connection. Bit 0 reserved Bit 1: 1st spindle Y-Delta target state. Bit 2: 2nd spindle Y-Delta target state. ... Bit 10: 10th spindle Y-Delta target state.	
R632	The adopted winring of Y-Delta of each spindle.	Each spindle Y-Delta adopted winring When the corresponding bit is 0, the spindle is Y connection; 1 means Delta connection. Bit 0 reserved Bit 1: 1st spindle Y-Delta target state. Bit 2: 2nd spindle Y-Delta target state. ... Bit 10: 10th spindle Y-Delta target state.	

R633~R636	1st~4th reference point (machine coordinate) arrival flag	The arrival flag for 1st~4th reference point (machine coordinate), will be updated at each PLC scan time. • R633: Represent arrival status of each axis for 1st reference point (machine origin). • R634: Represent arrival status of each axis for 2nd reference point (corresponding to Pr2801~). • R635: Represent arrival status of each axis for 3rd reference point (corresponding to Pr2821~). • R636: Represent arrival status of each axis for 4th reference point (corresponding to Pr2841~). Bit 0 reserved, not used. Bit 1 represents reference point arrival flag for 1st axis (X). If the axis has arrived then flag is 1 (On), otherwise 0 (Off). Bit 2 represents reference point arrival flag for 2nd axis (Y). If the axis has arrived then flag is 1 (On), otherwise 0 (Off). ... Bit 18 represents reference point arrival flag for 18th axis. If the axis has arrived then flag is 1 (On), otherwise 0 (Off). • To determine whether the arrival flag is on or not, the following conditions must be confirmed simultaneously. a. The homing process has finished (S Bit of Axis Home OK should be on) and the axis is not used as spindle. b. The axis has arrived at target machine position. c. The position error between servo command and motor feedback should be less or equals to Pr481~Pr500 Axial rapid travel in-position window⁹.	Read Only
R637~R639	Reserved for CNC	System Reserved Area	

⁹ <https://confluence.syntecclub.com.tw/display/TechManual/Pr481~Pr500+Axial+rapid+travel+in-position+window>

R641~ R658	Axis MPG command multiplier ratio	Set the manual ratio of each axis; set the manual multiplier ratio of the MPG JOG axis. R641: multiplier ratio of the 1st axis. R642: multiplier ratio of the 2nd axis. so on and so forth... The actual affective ratio is equal to the MPG segment (defined by R14, R22, R23) multiplied the multiplier ratio of the register. Register=1000 means the ratio is 1.0 Register=1 means the ratio is 0.001, but when Register=0 ratio is default as 1.0. Note: The R value only valid when the MPG corresponding coordinate is set up to be controlled by MLC (Ref. Parameter 2031~2040).	Write Only
R661~ R678	PLC axis program G00/ G01 feedrate override	Set the feedrate range of each axis PLC program -200~200, Default value 100. R661 → feedrate range of the 1st axis PLC program R662 → feedrate range of the 2nd axis PLC program so on and so forth...	Write Only
R681~ R698	PLC axis M code value	(please refer to R629)	Read Only
R700	Actual compound feedrate	Actual compound feedrate command, unit: LIU/min (PLC axis doesn't count in compound feedrate calculation) Always regarded as linear axis, for the rotary axis is in Imperial units, the value should be magnified 10 times. i.e. Pr17=2, under Imperial units, the 3rd axis is rotary axis feedrate F100. $R700 = 100 \text{ (IU/min)} * (10000 \text{ LIU/IU})$ LIU: 0.0001 inch, IU = inch $R703 = 100 \text{ (IU/min)} * (1000 \text{ BLU/IU})$ BLU: 0.001 deg, IU = deg Thus, R700 is 10 times to R703.	

R701~ R718	Axis Velocity	Velocity of each axis, Servo On mode: according to command value; Servo Off mode: according to feedback value, unit : BLU/min. R701 → single axis speed of the 1st axis R702 → single axis speed of the 2nd axis so on and so forth...	Read Only
R721~ R738	Program Coordinate	Axis program coordinate position, unit: minimum input unit (LIU). R721 → 1st axis program coordinate position R722 → 2nd axis program coordinate position so on and so forth... When control precision is micro meter and the value: 0.01 mm 0.01 deg, 0.0001 inch	Read Only
R741~ R758	Machine Coordinate	Axis machine coordinate position, unit: minimum input unit (LIU). R741 → 1st axis machine coordinate position R742 → 2nd axis machine coordinate position so on and so forth...	Read Only
R761~ R770	Spindle Machine Position	Display the machine position of the corresponding spindle, unit: 0.001 deg. R761 → 1st spindle machine position R762 → 2nd spindle machine position so on and so forth...	Read Only

R771~R780	Spindle Speed	Current Spindle Speed according to the feedback value, if there's no encoder then refer to the command value. If the spindle is not activated, the value should be fixed to 0. Unit: RPM. R771 → 1st spindle speed. R772 → 2nd spindle speed. so on and so forth...	Read Only
R782~R790	Spindle Gear Selection	Spindle Gear setting : 1 → 1st Gear 2 → 2nd Gear 3 → 3rd Gear 9 → Neutral R782 → 2nd spindle gear position R783 → 3rd spindle gear position so on and so forth... Note: 1st spindle gear position is controlled by R24 setting.	Write Only
R791~R794	Effective Spindle Selection	The machining spindle under multi-axis applications is selected by the content of the register, value definition: 1. the 1st spindle is the machining spindle 2. the 2nd spindle is the machining spindle 3. the 3rd spindle is the machining spindle 4. the 4th spindle is the machining spindle so on and so forth... R791 → machining spindle of 1st path R792 → machining spindle of 2nd path R793 → machining spindle of 3rd path R794 → machining spindle of 4th path When the spindle of the path is selected, the S command given to the path (\$1, \$2) will be directly send to the corresponding affective spindle. e.g. R791 = 1, R792 = 2, \$1 S1000 → 1000 RPM is assigned to 1st spindle \$2 S1500 → 1500 RPM is assigned to 2nd spindle Note: Before switching the machining spindle (R791~), if the current spindle is in the tapping state, must first cancel with G80 to avoid unexpected subsequent machining actions.	Write Only
R801~R830	MLC System Parameter	System parameter No.3421 ~ No.3450 of MLC parameter (Extension Parameter), for which one LADDER can be applicable to various machines, i.e. tool change...	Read Only

R831~R840	MLC System Parameter	System parameter No.3451 ~ No.3460 of MLC parameter (Extension Parameter), for which one LADDER can be applicable to various machines, i.e. tool change... ** These parameters don't need to Login and can be edit by user.	
R841~R844	Cause of Block Stop	Cause of block stop. 0: Normal. Stop by M00/M01 or Feedhold by user. 1: axis motion prohibited command, system automatically enter Feedhold status. R841 Cause of 1st path block stop. R842 Cause of 2nd path block stop. R843 Cause of 3rd path block stop. R844 Cause of 4th path block stop.	
~R850	Reserved for CNC	System reserved area	
R851~R860	Spindle Override	Override % of an individual spindle. R15 is the common override of all spindles, the actual override is R15*R851~. The unit of R851~ is %, and does not affected by Pr3207 setting. R851 1st Spindle override R852 2nd Spindle override R856 10th Spindle override	
R861~R870	Break point machine position	Machine coordinate position where machining stops. Unit is BLU, and the Registries support up to 10 axes. Machining Break Point Machine Position, unit: minimum length unit (BLU), same as the length parameter of system parameters. Registries support up to 10 axis.	

R871~ R879	Spindle Effective Speed Command	Spindle effective speed command, unit: RPM. Effective RPM means the actual speed command after calculation with spindle override or constant surface speed. The Effective Command is the final effective speed command value calculated after the spindle override command, constant surface speed cutting (G96/G97) and all other related factors. R871 1st Spindle Effective Speed Command R871 2nd Spindle Effective Speed Command ... R879 9th Spindle Effective Speed Command Note: Please refer to R930 for the 10th spindle.	Read Only
-----------------------	--	---	--------------

R880~R895	Reference Value for D/A of Servo Card or Torque Command	Control Mode	Machine Model	Descriptions	Write/Read
		Servo card D/A command (range: $\pm 10000\text{mV}$), effective version: 10.116.5	EZ, 10A	R881 is active as D/A command only if all the axis and spindle have not been defined as voltage control mode (i.e., Pr381~ $\neq$ 1, Pr1791~ $\neq$ 0/2).	
			10B/11B	R881 is active as D/A command only if all the axis and spindle which are linked to P5~P8 have not been defined as voltage control mode (i.e., Pr381~ $\neq$ 1, Pr1791~ $\neq$ 0/2)	
			Super-servo4	R880~R883 are active as servo card D/A command only if all the axis and spindle of the servo card have not been defined as voltage control mode (i.e., Pr381~ $\neq$ 1, Pr1791~ $\neq$ 0/2) R880 $\rightarrow$ 1 st servo card D/A command. R881 $\rightarrow$ 2 nd servo card D/A command. R882 $\rightarrow$ 3 rd servo card D/A command R883 $\rightarrow$ 4 th servo card D/A command	
			Super-servo6	R880 $\rightarrow$ 1 st servo card D/A command R881 $\rightarrow$ 2 nd servo card D/A command R882 $\rightarrow$ 3 rd servo card D/A command R883 $\rightarrow$ 4 th servo card D/A command	

Control Mode	Machine Model	Descriptions
	60A	Pre-defined zone for DAC value, reserve 16 channels. ps: Except Servo-6/60A, all other models needs to activate at least 1 axis to use this function.
Servo Card 1 st D/A & 2 nd D/A command (<u>Range $\pm 10000\text{mV}$</u>),	3, 6, 11, 21, eHMC series (<u>effective version: 10.114.56R, 10.116.00, 10.116.6G, 10.116.10A</u>)	Have 2 sets of DA (DA1 & DA2). 1. When DA is not occupied by any spindle or axis, Registries can control DA voltage. 2. Under condition 1, it's able to control DA1 output with R880; DA2 output with R884.
	FC-B, FC-60WA, F C-60CB and 60DB (<u>Valid version: 10.118.12R, 10.118.22</u>)	
	71SA (<u>Valid version: 10.118.12R, 10.118.22</u>)	Have 2 sets of D/A (DA1 & DA2) 1. When the DA is not occupied by any spindle or axis, Registries can control DA voltage. 2. Under condition 1, it's able to control DA1 output with R880; DA2 output with R884. 3. The DA output of FC-LSMK laser source is controlled by R881.
	70SB	The DA output of the laser source is controlled by R881.

		Control Mode	Machine Model	Descriptions	
			(Valid version: <u>10.118.12R,</u> <u>10.118.22)</u>		
		Expansion D/A Card on Multi-Axis Servo Drive Embedded Controller (Range:±10000, unit: mV)	Multi-Axis Servo Drive Embedded Controller (Valid version: <u>10.116.36I,</u> <u>10.116.41,</u> <u>10.117.41)</u>	When Pr3261~Pr3263 =n004 (n: card station number), R880/R884/R888/R892 represents the 1st~4th DA port on expansion card.	
		Rated Torque Reference Command (Unit: Rated torque/1000)	6MD, 20, 21, 200 series (Valid version: <u>10.114.56, 10.116.0)</u>	Serial controllers support torque control. Use R627 to enable/disable torque control mode. When axis under R627 torque control mode, these registers are torque reference command, with positive value means positive output, negative value means negative output. R880 → 1 st axis torque command. R881 → 2 nd axis torque command. ... R895 → 16 th axis torque command Note1: Not Support the 17 th & 18 th axis. Note2: The control interface only supports version 10.118.10 and below.	
R896~R899	A/D value	(1) A/D converter value, 2 channel per servo 6 board. (2) Pre-defined zone for ADC value, reserve 4 channels.			Read Only

R900	Data Acquisition Trigger (On DOS System) Ballbar Adjust Trigger (On WinCE)	A/D converter value, 2 channel per servo 6 board (DOS only) DOS system: When this R value is On, the oscilloscope will activate the capture function. WinCE system: When this R value is On, start executing ballbar tuning. When this R value is Off, stop executing ballbar tuning.	Write Only
R901~R918	Axis temperature compensation amount	Temperature Compensation Amount, unit: BLU R901 1 st axis temperature compensation amount R902 2 nd axis temperature compensation amount ... R918 18 th axis temperature compensation amount	Write Only
R920	EDM discharge enable.	EDM discharge enabled. (0: Disable, 1: Enable)	Write Only
R921	EDM control mode	EDM discharge control mode. 0: Fixed Frequency. suitable for laser machine. 1: Fixed Energy. suitable for EDM.	Write Only
R922	EDM normal ON time	EDM normal ON time level, unit: Discharge resolution set by Pr3831. (When PR3261~Pr3263= n004, this parameter means the setting for PWM on Expandable Card of Servo Drive Embedded Controller. Valid version 10.116.36I , 10.116.41 , 10.117.41)	Write Only
R923	EDM normal OFF time	EDM normal OFF time level, unit: Discharge resolution set by Pr3831. (When PR3261~Pr3263= n004, this parameter means the setting for PWM on Expandable Card of Servo Drive Embedded Controller. Valid version 10.116.36I , 10.116.41 , 10.117.41)	Write Only

R924	EDM arc ON time	EDM arc ON time level, unit: Discharge resolution set by Pr3831.	Write Only
R925	EDM arc OFF time	EDM arc OFF time level, unit: Discharge resolution set by Pr3831.	Write Only
R926	Reserved	Reserved for system.	
R927	Total Discharge Count	Constant energy machining, total discharge count.	Read Only
R928	Normal Discharge Count	Constant energy machining, total normal discharge count.	Read Only
R929	Abnormal Discharge Count	Constant energy machining, total abnormal discharge count.	Read Only
R930	Spindle Effective Speed Command	The effective speed command of the 10 th spindle, unit: RPM. The Effective Command is the final effective speed command value of spindle command calculation result from percentage, constant linear velocity control on surface (G96/G97), and all related factors.	
R934	MPG: 1: x1 2: x10 3: x100 4: x magnified based on Pr2001 setting.	4th MPG override selection.	Write only

R935	MPG: 1: x1 2: x10 3: x100 4: x magnified based on Pr2001 setting.	5th MPG override selection.	Write only
R936	MPG: 1: x1 2: x10 3: x100 4: x magnified based on Pr2001 setting.	6th MPG override selection.	Write only
R937	MPG: 1: x1 2: x10 3: x100 4: x magnified based on Pr2001 setting.	7th MPG override selection.	Write only
R938	Start program No. of "auxiliary program function before and after machining"	Start program No. of "auxiliary program function before and after machining" 1. When value = or <0, auxiliary program will not be executed. 2. Auxiliary program needs to start with letter "O". (O####) 3. If value is less than 4 digit, system recognizes it as a 4 digit program No.; if less than 6 digit, system recognizes a 6 digits. Example: • R938 = 12 → Program No. O0012 • R938 = 1234 → Program No. O1234 • R938 = 12345 → Program No. O012345 • R938 = 123456 → Program No. O123456	Read/Write

R939	End program No. of "auxiliary program function before and after machining"	End program No. of "auxiliary program function before and after machining" 1. When value = or <0, auxiliary program will not be executed. 2. Auxiliary program needs to start with letter "O". (O####) 3. If value is less than 4 digit, system recognizes it as a 4 digit program No.; if less than 6 digit, system recognizes a 6 digits. i.e.: • R939 = 12 → Program No. O0012 • R939 = 1234 → Program No. O1234 • R939 = 12345 → Program No. O012345 • R939 = 123456 → Program No. O123456	Read/Write
R940	Excessive Displacement During Power Off	In absolute system, when CNC axis position displacement exceeds the range set in Pr1021~1040 during power off, the Bit of corresponding R value will be On. When C84 ON this R value is cleared to 0. If needs the Bit value to send MLC alarms of corresponding axis, it can be processed by filling the value into R44~R49. Bit 0: 1st axis excess displacement during power off Bit 1: 2nd axis excess displacement during power off ... Bit 18: 18th axis excess displacement during power off Note: Due to the limitation of ladder file format, the R value that can be accessed on the ladder edit is only till R639. Therefore, when using this R value, better choose a user-defined section and an accessible R value then write the ladder as the example shows below (take R50 for example). ? 未知的附加檔案	Read Only
R941~R950	PLC Spindle Command	PLC Spindle Command, unit: RPM. This function enabled by R580. R941 → 1st spindle R942 → 2nd spindle ... R950 → 10th spindle	Read/Write

R951~R954	Current interpolation mode	Current interpolation mode, possible value: 1/2/3/4/33/34/35 R951 current interpolation mode of 1st path. R952 current interpolation mode of 2nd path. R953 current interpolation mode of 3rd path. R954 current interpolation mode of 4th path.	Read Only
R956~R959	Extended Stroke Limit	Switch & Operation Status of extended stroke limit The extended limit switch is defined by the Bit value, Bit =1 means activated; =0 means de-activated. R956 .0~.31 : 01~32 set of extended stroke limit switch. R957 .0~.31 : 01~32 set of extended stroke limit operation status. R958 .0~.31: 33~64 set of extended stroke limit switch. R959 .0~.31: 33~64 set of extended stroke limit operation status. Notification: 1. Due to the limitation of ladder file format, the R value that can be accessed on the ladder edit is only till R639. Therefore, when using this R value, better choose a user-defined section and an accessible R value then write the ladder as the example shows below (take R50 for example). ? 未知的附加檔案 2. The R value only supports 81RG series for now.	Read/Write
R961~R978	Home Dog Shift	The shift (offset) from Home Dog signal to motor index. Unit for rotary encoder: 0.01 rev; for linear encoder: 0.01 mm. (Equivalent to System Data 56~59) R961 is the home dog shift limit of the 1st axis R962 is the home dog shift limit of 2nd axis so on and so forth.....	

R981~R998	Axis Absolute Counter	The absolute position feedback of each axis, which is the axial motor encoder feedback. For axis with dual feedback control, the Registry displays feedback from 2nd (external) encoder. Unit: BLU. R981, the position sensor value of the 1st axis R982, the position sensor value of the 2nd axis so on and so forth.....	Read Only
R1000	Part count	Part count	Read/Write
R1002	Required part count	Required part count	Read/Write
R1004	Total part count	Total part count	Read/Write

R1005~R1007	Controller S/ N ASCII Code	4 bytes as one set, uses “little endian” display method; below is an example with M9A0001: <table><tr><th>Number</th><th>Hexadecimal</th><th>Decimal (Controller Display)</th></tr><tr><td>R1005</td><td>0x30 0x41 0x39 0x4D 0 A 9 M</td><td>809580877</td></tr><tr><td>R1006</td><td>0x31 0x30 0x30 1 0 0</td><td>3223600</td></tr><tr><td>R1007</td><td>0</td><td>0</td></tr></table> R1007 is reserved for “controller serial number” expansion. Situation Applied: If customers have encryption demands for “machine MACRO”, usually check the controller serial numbers with CHKSN ("Serial Number") inside the MACRO. But if customer wants to protect the serial number again with the encryption logic designed by themselves, then will decrypt inside the MACRO to compare the controller serial "number", and R1005~R1007 can provide customers to perform this checkup. Note: For “little endian” method, the least significant bit is in front of the most significant bit.	Number	Hexadecimal	Decimal (Controller Display)	R1005	0x30 0x41 0x39 0x4D 0 A 9 M	809580877	R1006	0x31 0x30 0x30 1 0 0	3223600	R1007	0	0	Read only
Number	Hexadecimal	Decimal (Controller Display)													
R1005	0x30 0x41 0x39 0x4D 0 A 9 M	809580877													
R1006	0x31 0x30 0x30 1 0 0	3223600													
R1007	0	0													
R1009	Motion interpolation times after power-on	Motion interpolation times after power-on, refer to System Data No. 4, multiply the 2 factors can get the run time after power-on. i.e. R1009=1000, and System Data No. 4=3000, means that the controller has been powered-on for 3 seconds.	Read only												
R1010	Cutting time per cycle	Machining time per cycle, unit: second. Not retained after power-off.	Read only												
R1011	Accumulate cutting time	Accumulating machining time, unit: second. Value reserved when power off, value can be reset to 0 by user on HMI manually.	Read only												
R1012	Power-on time_1	Power-on time, unit: second. Value not-reserved after power off.	Read only												

R1013	Accumulate power-on time	Accumulated power-on time, unit: second. Value reserved after power-off.	Read only
R1014	Installation cutting time	Installation cutting time for total, unit: hour. Value reserved after power-off, can only be cleared at the 1st installation.	Read only
R1015	Power-on time_2	Power-on time, unit: hour. Value not-reserved after power-off.	Read only
R1016	System date	YYYYMMDD YYYY: Year; MM: Month; DD: Day	Read only
R1017	System time	HHMMSS HH: Hour; MM: Minute; SS: Second	Read only
R1018	Cutting time per cycle	Cutting time of last machining part, update after a part (program) is finished, unit: ms. value not-reserved after power-off.	Read only
R1019	PLC Version	PLC Version, format is ##,##. R1019=101 means V1,01 R1019=1020 means V10.20.	Read/Write
R1020~R1023	PLC File System	R1020 allowed file number of PLC file system. R1021 set store data width, unit is number of register. R1022 set store data height, unit row number. R1023 set the starting address of file system. Control format as follow:	Write only
R1024~R1032	User define/ System use	Mill: Used for auto tool center & tool height measure function; X Axis: R1024~R1026 Y Axis: R1027~R1029 Z Axis: R1030~R1032 Lathe: User-Defined Registers	Read/Write

R2050~R2090	User define/ System use	When Pr3810 Paralleled executing multiple M code in one block is set to 0, these registers are user defined; when Pr3810 is set to 1, the registers in this section are used by system, and defined as below:			Read only
		Group 1	M Code Read	S29,S11,S12,S13,S14	
			M Code(00~99)	R2050(R1) R2051 R2052 R2053 R2054	
			M Code Finish	C38	
		Group 2	M Code Read	R615.24(R615.2) R615.24 R615.25 R615.26 R615.27 R615.28	
			M Code(00~99)	R2060(R616) R2061 R2062 R2063 R2064	
			M Code Finish	R615.0	
		Group 3	M Code Read	R619.24(R619.2) R619.24 R619.25 R619.26 R619.27 R619.28	
			M Code(00~99)	R2070(R620) R2071 R2072 R2073 R2074	
			M Code Finish	R619.0	
		Group 4	M Code Read	R623.24(R623.2) R623.24 R623.25 R623.26 R623.27 R623.28	

		<table><tr><td>M Code(00~99)</td><td>R2080(R624) R2081 R2082 R2083 R2084</td></tr><tr><td>M Code Finish</td><td>R623.0</td></tr></table>	M Code(00~99)	R2080(R624) R2081 R2082 R2083 R2084	M Code Finish	R623.0	
M Code(00~99)	R2080(R624) R2081 R2082 R2083 R2084						
M Code Finish	R623.0						
R2091~R2099	User define	User defined	Read/Write				
R2100~R3100	User define/ System use	When Pr3228 'Enable tool manager' =0, these registers are user-defined; when Pr3228=1, registers are used by HMI, and HMI system will auto load or save to Registry Table 2100~3100.	Read/Write				
R3101~R4095	User define	User defined	Read/Write				

R4096~R4099	Accumulate look-ahead length	G01, G02, G03 look-ahead accumulated length (unit : mm) (Effective version: 10.118.12 and above) R4096 : 1st path accumulated length R4097 : 2nd path accumulated length R4098 : 3rd path accumulated length R4099 : 4th path accumulated length Note 1 : Maximum Accumulated length is 2147483647 (unit: mm) Note 2 : Stop to accumulate length under Feedhold and Emergency stop. Note 3 : After reset and Emergency stop, clear the accumulator. Note 4 : M30 will clear the accumulator Note 5 : When executing the M Code of Workpiece Counter(Pr3804), clear clear the accumulator of all paths. Note 6 : Using M99 to repeat cycleing, the accumulated length will not be clear. Unless Pr3804 is 99 (Note 5). Note 7 : Special Situation (Insert Program、 Tool break retract、 Feedhold Point Retract 、 Tool Retraction) 1. Insert Program : Continue to accumulate when executing sub-program. 2. Tool break retract : Accumulator only starts on the middle of the program, user can use #1048, #1052 to evaluate remain time. 3. Feedhold Point Retract : Stop to accumulate under Feedhold. Restart to accumulate after cycling start. The length of tool back will not accumulate. 4. Tool Retraction : Tool Retraction (that is, the moment between Tool retract and recover) will not accumulate length.	
R4100~R5000	Reserved for CNC	System reserved area	

R5001~R5018	Axis Following Error	R5001: 1st axis following error. R5002: 2nd axis following error. so on and so forth..	Read only
R5021~R5024	Spindle Max Cutting Speed	Spindle maximum cutting speed R5021 is the spindle of 1st path. R5022 is the spindle of 2nd path. R5023 is the spindle of 3rd path. R5024 is the spindle of 4th path.	Read only
R5025~R5028	Reserved for CNC	System reserved area	
R5029	Modbus Slave Error Code	Modbus Slave Error Code 0 : Slave (Server) activate success 1 : Slave (Server) activate failure 2 : Invalid Port 3 : Invalid Station ID 4 : Incorrect packet length 5 : Incorrect packet content 6 : Data confirmation error (CRC error) 7 : Machining file upload abort command not received 8 : Machining file read error 9 : Response return failure	Read only

R5030

Modbus Protocol Setup

When R5030.0 is set to disabled, PLC Modbus component communication use the set value of Pr39xx (COM2: Pr3941~Pr3950, COM3: Pr3961~ Pr3970)

Write/Read

b ₀	Enable Modbus protocol setting	Disable: 0 Enabled: 1
b ₁	Data length	7 bit: 0 8 bit: 1
b ₃ b ₂	Polarity Bit	No: 00 Odd Corresponding: 01 Even Corresponding: 11
b ₄	Stop Bit	1 bit: 0 2 bit: 1
b ₈ b ₇ b ₆ b ₅	Baud Rate	2400 bps : 0000 4800 bps : 0001 9600 bps : 0010 19200 bps : 0011 38400 bps : 0100 57600 bps : 0101 115200 bps : 0110 230400 bps : 0111 460800 bps : 1000 912600 bps : 1001
b ₉ ~b ₃₁	System reserved area (do not use)	

		Communication Format R5030 setting example: Baud Rate "9600"/Stop bit "1"/Polarity Bit "None"/Data length "8"/Enable "1" $b_8b_7b_6b_5=0010$ / $b_4=0$ / $b_3b_2=00$ / $b_1=1$ / $b_0=1$ $R5030=001000011_{(2)} \rightarrow$ Binary format should be converted to decimal format, $R5030=67_{(10)}$	
R5031	1 st Serial MPG Pulse Count	Absolute value of Serial MPG Pulse Count, corresponds to M3 IO 1st station MPG pulse feedback signal. After software version 10.118.13, when activating the customized IO chart (Pr5 *I_O board type¹⁰=100), it's able to modify the corresponding initial point R at the I/O chart page and not using R5031~R5033.	Read Only
R5032	2 nd Serial MPG Pulse Count	As above, corresponds to MPG pulse feedback signal of M3 IO 2 nd station.	Read Only
R5033	3 rd Serial MPG Pulse Count	As above, corresponds to MPG pulse feedback signal of M3 IO 3 rd station.	Read Only
R5034~R5038	Reserved for Serial MPG	Serial MPG Function Reserved Area	

¹⁰ https://confluence.syntecclub.com.tw/display/TechManual/Pr5+I_O+board+type

R5039	Modbus Error Code	Modbus Error Code Instruction	
		R5039	Detail
		0	Success
		1	Function Code error
		2	Address error
		3	Data Value error - Data value is defined invalid by Slave
		4	Slave Device error
		5	Confirm Acknowledge (ACK), when Slave defined longer process time is needed, then return ACK to prevent timeout.
		6	Slave Device Busy
		8	Memory Parity error
		10	Gateway Path Unavailable
		11	Gateway target Device failed to Respond
		245	Non-supported Function Code
		246	Modbus Packet format error
		247	The COM Port assigned by R5040 is incorrect.
		248	Server busy, halt communication command.

		<table><tr><th>R5039</th><th>Detail</th></tr><tr><td>249</td><td>Data confirmation error (CRC error)</td></tr><tr><td>250</td><td>TimeOut error</td></tr><tr><td>251</td><td>Read COM Port error</td></tr><tr><td>252</td><td>Write COM Port error</td></tr><tr><td>253</td><td>Open COM Port error</td></tr><tr><td>254</td><td>Unknown error</td></tr></table>	R5039	Detail	249	Data confirmation error (CRC error)	250	TimeOut error	251	Read COM Port error	252	Write COM Port error	253	Open COM Port error	254	Unknown error	
R5039	Detail																
249	Data confirmation error (CRC error)																
250	TimeOut error																
251	Read COM Port error																
252	Write COM Port error																
253	Open COM Port error																
254	Unknown error																
R5040	Modbus COM Port	Modbus COM Port is used to set up the actual wirings. If the hardware is SUPER/10/20/200, the COM Port is 3 (R5040=3); if it's eHMC/3/6/11/21, the COM Port is 2 (R5040=2). COM Port is Modbus TCP when set to 9, it would be able to transmit data through the Internet (LAN), the specifications are the same as Modbus RS485, and differences are: 1. Controller client (Master) and the connecting device server (Slave) should be under the same domain. 2. The station ID of the connecting device server (Slave) should set to be the same as the 4th digit of IP. i.e. If the IP is 192.168.1.5, the station ID should set to be 5. 3. Can connect at most 2 server (Slave) devices. 4. The communication error timeout can be set in Pr 3971.															
R5041~R5060	Reserved for CNC	Syntec Reserved Area															
R5061~R5080	A/D value	When Pr3261~Pr3263 =n004 (n: card station number), R5061~R5066 mapping to 1st~6th DA port on expansion card of Syntec Servo Drive Embedded Controller. <u>(Effective version: 10.116.36I, 10.116.41, 10.117.41)</u>	Read only														

R5081~R5100	Thermal value	Pre-defined zone for Thermal value, reserve 20 channels	Read only
R5101~R5118	Axis Index Counter	R5101: 1 st axis index counter R5102: 2 nd axis index counter so on and so forth...	Read only
R5121~R5138	Accumulated Total Index Signal Counts	Axis corresponding register value +1 whenever an index signal is read. R5121: 1 st axis total index counts R5122: 2 nd axis total index counts so on and so forth...	Read only
R5139~R5140	Reserved for CNC	Reserved for system	
R5141~R5150	Spindle Torque Load	Spindle Torque Load: actual torque output of motor, unit % of rated torque. R5141: 1 st spindle torque load R5142: 2 nd spindle torque load ... R5150: 10 th spindle torque load Note 1: The function only works for serial bus communication. Note 2: Supporting Version 10.116.9 and above. Note 3: For the supporting versions and brands of EtherCAT serial, Refer to EtherCAT驱动器应用手册#主轴与轴向负载率 ¹¹ . Note 4: For Panasonic RTEX (A6N) set driver parameter Pr7.34=18, supporting versions refer to RTEX驱动器应用手册 ¹² . Note 5: When spindle type is inverter spindle (Pr1791~ = 0, 5), this value will be 0.	Read only

¹¹ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=175063716#EtherCAT驱动器应用手册-主轴与轴向负载率>

¹² <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=261391560>

R5151~R5168	Axial Torque Load	Axial Torque Load: actual torque output of motor, unit % of rated torque. R5151: 1st axis (X) torque load R5152: 2nd axis (Y) torque load so on and so forth... Note 1: The function only works for serial bus communication. Note 2: Supporting Version 10.116.15 and above. Note 3: Please refer to EtherCat driver application manual for the supporting versions and brands of EtherCat serial PLC. Note 4: For Panasonic RTEX (A6N) set driver parameter Pr7.34=18, supporting versions refer to RTEX驱动器应用手册¹³	Read
R5171~R5188	Axial Rated Torque Limit Per mille	Torque limit ratio, unit thousands (1/1000) of rated torque. The Registers =0 means the axis is under maximum torque for position control. R5171: 1st axis (X) torque limit ratio R5172: 2nd axis (Y) torque limit ratio so on and so forth.. Note 1: The function only supports 1. YASKAWAM2/M3 2. SYNTECM3 3. EtherCAT serial supporting versions and brand, ref. EtherCAT驱动器应用手册¹⁴ Note 2: To activate the function, enable the torque limitation function by setting driver parameters and restart the driver. Note 3: If R5171~5188 are not 0, both Pulse and Serial System do not protect the loss pulse and tracking error. Note 4: Supporting version: 10.116.20, 10.117.21 and above Note 5: Current supporting modes are NC Axis Position Mode, Spindle Orientation Mode (C61), Spindle Speed Mode (C64), Spindle C Axis Mode (C63)	Write/ Read

13 <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=261391560>

14 <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=175063716>

R5189~R5190	Reserved for CNC	Reserved for system	
R5191~R5208	Velocity FeedForward under Axis Position Mode	Velocity FeedForward under Axis Position Mode, unit: um/sec. R5191 is the 1st axis Velocity FeedForward under Axis Position Mode. R5192 is the 2nd axis Velocity FeedForward under Axis Position Mode. so on and so forth..... Note1: The function only supports YASKAWA M3 and Panasonic RTEX. Note2: Loss pulse and tracking error won't be protected when using this function. Note3: YASKAWA M3 supporting version should be 10.116.51, 10.117.51 and above. Note4: Panasonic RTEX supporting version should be 10.118.12I, 10.118.17 and above, please refer to RTEX Driver Application Manual(CHAT)¹⁵ for related parameters settings of RTEX velocity feedforward.	Read/Write

¹⁵ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=296273996>

R5291	2 nd Driver IO Signal, default as external latch signal (External latch 1 input)	It's able to monitor other driver signals (Torque Limit or Brake signal) by Pr3266~Pr3267 Select the first set of driver signal to be monitored¹⁶. Pr3267 = 1, R5291.1~R5291.18 corresponds to the torque arrival signal of the 1st~18th axis, the Bit becomes 1 when the torque arrives the limit value. Pr3267 = 2, R5291.1~R5291.18 corresponds to the EXT1 signal of the 1st~18th axis, the Bit becomes 1 when the driver receives the EXT1 input signal. Pr3267 = 3, R5291.1~R5291.18 corresponds to the Brake signal of the 1st~18th axis, the Bit becomes 1 when the driver activates the brake. When applying high speed function G31, it would be able to monitor the signal with R value and adjust it more easily if the signal resource of G31 is set up to come from driver EXT1. Note1: Supporting Version 10.116.36Y, 10.116.51, 10.117.51 and above Note2: Please refer to R30 for the signal state supported by drivers from each brand.	Read Only
R5292~R5879	Reserved for CNC	System Reserved Area	
R5880	Cruise Control Switch	Axis Switch between Positioning Control Mode and Gap Control Mode 0: Switch back to Positioning Control Mode 1: Switch the 1st axis to Gap Control Mode 2: Switch the 2nd axis to Gap Control Mode ... so on and so forth.	
R5881	Target Capacitance	Cruise Control Mode Target Capacitance (count: 0~500,000)	
R5882	Tracing Slope	Cruise Control Mode Tracing Slope (pulse/count: -500,000~500,000)	

¹⁶ <https://confluence.syntecclub.com.tw/display/TechManual/Pr3266~Pr3267+Select+the+first+set+of+driver+signal+to+be+monitored>

R5883	Check Window / Max Capacitance	~10.118.28M, 10.118.36: Cruise Control Mode Position Check Window (BLU: 0~10,000) 10.118.28N, 10.118.37 and above: Max Capacitance(count: -32,768~32,767), 1 Unit equals 1000 Units of R5881; Drive version must be 2.15.0, 2.14.105, 3.0.0 and above.	
R5884	Capacitance Threshold / Min Capacitance	~10.118.28M, 10.118.36:Capacitance Threshold Value (count: 0~500,000) 10.118.28N, 10.118.37 and above: Min Capacitance(count: -32,768~32,767), 1 Unit equals 1000 Units of R5881; Drive version must be 2.15.0, 2.14.105, 3.0.0 and above.	
R5885	Cruise Control Status	Cruise Control Axial Status 0: Not In Cruise Control Mode 1: In Cruise Control Mode	
R5886	Cruise in Position	Cruise Control Status 0: Cruise control unfinished/not support 1: Cruise control finished	
R5887	Increment Position Limit	Increment Position Limit (BLU: -999,999~999,999)	
R5888~R5890	Reserved for CNC	System Reserved Area	
R5891~R5894	Coordinate Feedrate Feedback	Combined feedrate FEEDBACK of axis group, unit LIU/min. R5891: Combined feedrate of the 1st path R5892: Combined feedrate of the 2nd path R5893: Combined feedrate of the 3rd path R5894: Combined feedrate of the 4th path	Read
R5895~R5900	Reserved for CNC	Reserved for system	

R5901~R5904	Coordinate Feedrate Command	Combined feedrate COMMAND of axis group, unit: LIU/min. R5901: Combined feedrate of the 1st path R5902: Combined feedrate of the 2nd path R5903: Combined feedrate of the 3rd path R5904: Combined feedrate of the 4th path	Read
R5905~R5910	Reserved for CNC	System Reserved Area	
R5911~R5930	Robot Coordinate Feedrate Command	Robot Path Actual Speed Command Value, unit: LIU/s R5911: Shows the maximum single axis speed percentage the internal axis of the 1st path can reach (unit: 0.001%) R5912: Shows the linear direction feeding speed of the 1st path end point. R5913: Shows the rotary direction feeding speed of the 1st path end point. R5914: Shows the maximum single axis speed percentage the external axis of the 1st path can reach (unit: 0.001%). R5915: Reserved. Each path provides 5 R values, and so on to R5930.	Read
R5931~5996	Reserved for CNC	System Reserved Area	
R5997	Preview Manager Classification	9: Syntec customer service staff 99: Machine Manufacturer 100: End-User Manager 101: End-User Operator 999: Not Logged In Effective version 10.116.10 and above	Read
R5998	Background Action	Register for PLC customized Image ACTION	Read/Write

R5999	Tapping Maximum Tracking Error	Max. Z axis to spindle following error under rigid/rapid tapping	Read
R6000~R7999	F-RAM memory for user define	F-RAM memory for user self defined	Read/Write
R8000~R8500	MLC System Parameter	R8001~R8500 correspond to System parameter No.5001 ~ No.5500.	Read
R8500~R8999	Reserved for CNC	Reserved for system	

R9000~R9999	For robot kernel	Robot Kernel Operating Area (Not Customized Area)	Read / Write
R10000~R10499	Reserved for Vision Memory	Reserved for Vision system, ref. 3.7 system variables (R/L) instructions ¹⁷	Read
R10500~R14999	Reserved for CNC	Reserved for system	
R15000~R15199	Reserved for Vision Processing	Reserved for Vision system, ref. 3.7 system variables (R/L) instructions ¹⁸	
R15200~R15499	Reserved for Robot HMI	Reserved for Robot HMI	
R15500~R15999	Reserved for Vision	Reserved for Vision system, ref. 3.7 system variables (R/L) instructions ¹⁹	

¹⁷ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=94838084>

¹⁸ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=94838084>

¹⁹ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=94838084>

R16000	Trigger Software Panel _CycleStart Action	The register represents _CycleStart action on software panel. When user triggers _CycleStart on software panel, this register = 1.	Read
R16001	Disable Direct Cycle Start by Software Panel	If this Register = 0 , triggering _CycleStart on software panel will start running program ; if this Register ≠ 0 , system will not start when triggering _CycleStart .	Write/Read
R16002	Trigger Software Panel _FeedHold Action	The register represents _FeedHold action on software panel. When user triggers _FeedHold on software panel, this register = 1.	Read
R16003	Disable Direct Feed Hold by Software Panel	If this Register = 0 , triggering _FeedHold on software panel will pause running program ; if this Register ≠ 0 , system will not pause when triggering _FeedHold .	Write/Read
R16004	Trigger Software Panel _Jog+Axis_1, _StopJog+Axis_1 Action	The register represents _Jog+Axis_1, _StopJog+Axis_1 action on software panel. When user triggers _Jog+Axis_1 on software panel, this register = 1. When user triggers _StopJog+Axis_1 on software panel, this register = 0.	Read
R16005	Disable Direct X Axis JOG + by Software Panel	If this Register = 0 , triggering _Jog+Axis_1 on software panel, system will do X Axis JOG + ; if this Register ≠ 0 , system will not do X Axis JOG + when triggering _Jog+Axis_1 .	Write/Read

R16006	Trigger Software Panel _Jog-Axis_1, _StopJog-Axis_1 Action	The register represents _Jog-Axis_1, _StopJog-Axis_1 action on software panel. When user triggers _Jog-Axis_1 on software panel, this register = 1. When user triggers _StopJog-Axis_1 on software panel, this register = 0.	Read
R16007	Disable Direct X Axis JOG - by Software Panel	If this Register = 0, triggering _Jog-Axis_1 on software panel, system will do X Axis JOG - ; if this Register ≠ 0, system will not do X Axis JOG - when triggering _Jog-Axis_1.	Write/ Read
R16008	Trigger Software Panel _Jog+Axis_2, _StopJog+Axis_2 Action	The register represents _Jog+Axis_2, _StopJog+Axis_2 action on software panel. When user triggers _Jog+Axis_2 on software panel, this register = 1. When user triggers _StopJog+Axis_2 on software panel, this register = 0.	Read
R16009	Disable Direct Y Axis JOG + by Software Panel	If this Register = 0, triggering _Jog+Axis_2 on software panel, system will do Y Axis JOG + ; if this Register ≠ 0, system will not do Y Axis JOG + when triggering _Jog+Axis_2.	Write/ Read
R16010	Trigger Software Panel _Jog-Axis_2, _StopJog-Axis_2 Action	The register represents _Jog-Axis_2, _StopJog-Axis_2 action on software panel. When user triggers _Jog-Axis_2 on software panel, this register = 1. When user triggers _StopJog-Axis_2 on software panel, this register = 0.	Read

R16011	Disable Direct Y Axis JOG - by Software Panel	If this Register = 0, triggering _Jog-Axis_2 on software panel, system will do Y Axis JOG - ; if this Register ≠ 0, system will not do Y Axis JOG - when triggering _Jog-Axis_2.	Write/Read
R16012	Trigger Software Panel _Jog+Axis_3, _StopJog+Axis_3 Action	The register represents _Jog+Axis_3, _StopJog+Axis_3 action on software panel. When user triggers _Jog+Axis_3 on software panel, this register = 1. When user triggers _StopJog+Axis_3 on software panel, this register = 0.	Read
R16013	Disable Direct Z Axis JOG + by Software Panel	If this Register = 0, triggering _Jog+Axis_3 on software panel, system will do Z Axis JOG + ; if this Register ≠ 0, system will not do Z Axis JOG + when triggering _Jog+Axis_3.	Write/Read
R16014	Trigger Software Panel _Jog-Axis_3, _StopJog-Axis_3 Action	The register represents _Jog-Axis_3, _StopJog-Axis_3 action on software panel. When user triggers _Jog-Axis_3 on software panel, this register = 1. When user triggers _StopJog-Axis_3 on software panel, this register = 0.	Read
R16015	Disable Direct Z Axis JOG - by Software Panel	If this Register = 0, triggering _Jog-Axis_3 on software panel, system will do Z Axis JOG - ; if this Register ≠ 0, system will not do Z Axis JOG - when triggering _Jog-Axis_3.	Write/Read

R16016~ R16067	Trigger Software Panel _Jog+/- Axis_4~16 , _StopJog+/- Axis_4~16 Action	Same as above	Read
	Disable Direct 4th~16th Axis JOG + / - by Software Panel		Write/ Read
R16068	Trigger Software Panel _MPGSimuOn , _MPGSimuOff Action	The register represents _MPGSimuOn, _MPGSimuOff action on software panel. When user triggers _MPGSimuOn on software panel, this register = 1. When user triggers _MPGSimuOff on software panel, this register = 0.	Read
R16069	Disable Direct MPG Simulation by Software Panel	If this Register = 0 , triggering _MPGSimuOn on software panel, system will enable MPG Simulation ; if this Register ≠ 0 , system will not enable MPG Simulation when triggering _MPGSimuOn .	Write/ Read
R16070	Trigger Software Panel _SetMachinePosAxis_1 Action	The register represents _SetMachinePosAxis_1 action on software panel. When user triggers _SetMachinePosAxis_1 on software panel, this register = 1.	Read

R16071	Disable Direct Set X Axis Machine Position by Software Panel	If this Register = 0 , triggering _SetMachinePosAxis_1 on software panel, system will Set X Axis Machine Position ; if this Register ≠ 0 , system will not Set X Axis Machine Position when triggering _SetMachinePosAxis_1 .	Write/ Read
R16072	Trigger Software Panel _SetMachinePosAxis_2 Action	The register represents _SetMachinePosAxis_2 action on software panel. When user triggers _SetMachinePosAxis_2 on software panel, this register = 1.	Read
R16073	Disable Direct Set Y Axis Machine Position by Software Panel	If this Register = 0 , triggering _SetMachinePosAxis_2 on software panel, system will Set Y Axis Machine Position ; if this Register ≠ 0 , system will not Set Y Axis Machine Position when triggering _SetMachinePosAxis_2 .	Write/ Read
R16074	Trigger Software Panel _SetMachinePosAxis_3 Action	The register represents _SetMachinePosAxis_3 action on software panel. When user triggers _SetMachinePosAxis_3 on software panel, this register = 1.	Read
R16075	Disable Direct Set Z Axis Machine Position by Software Panel	If this Register = 0 , triggering _SetMachinePosAxis_3 on software panel, system will Set Z Axis Machine Position ; if this Register ≠ 0 , system will not Set Z Axis Machine Position when triggering _SetMachinePosAxis_3 .	Write/ Read
R16076~ R16101	Trigger Software Panel _SetMachinePosAxis_4 ~16th Action	Same as above	Read

	Disable Direct Set 4th~16th Axis Machine Position by Software Panel		Write/Read
R16102	Trigger Software Panel _ManualControlOnAxis_1 , _ManualControlOffAxis_1 Action	The register represents _ManualControlOnAxis_1, _ManualControlOffAxis_1 action on software panel. When user triggers _ManualControlOnAxis_1 on software panel, this register = 1. When user triggers _ManualControlOffAxis_1 on software panel, this register = 0.	Read
R16103	Disable Direct X Axis MANUAL CONTROL by Software Panel	If this Register = 0, triggering _ManualControlOnAxis_1 on software panel, system will enable X Axis MANUAL CONTROL ; if this Register ≠ 0, system will not enable X Axis MANUAL CONTROL when triggering _ManualControlOnAxis_1.	Write/Read
R16104	Trigger Software Panel _ManualControlOnAxis_2 , _ManualControlOffAxis_2 Action	The register represents _ManualControlOnAxis_2, _ManualControlOffAxis_2 action on software panel. When user triggers _ManualControlOnAxis_2 on software panel, this register = 1. When user triggers _ManualControlOffAxis_2 on software panel, this register = 0.	Read

R16105	Disable Direct Y Axis MANUAL CONTROL by Software Panel	If this Register = 0, triggering _ManualControlOnAxis_2 on software panel, system will enable Y Axis MANUAL CONTROL ; if this Register ≠ 0, system will not enable Y Axis MANUAL CONTROL when triggering _ManualControlOnAxis_2.	Write/Read
R16106	Trigger Software Panel _ManualControlOnAxis_3 , _ManualControlOffAxis_3 Action	The register represents _ManualControlOnAxis_3 , _ManualControlOffAxis_3 action on software panel. When user triggers _ManualControlOnAxis_3 on software panel, this register = 1. When user triggers _ManualControlOffAxis_3 on software panel, this register = 0.	Read
R16107	Disable Direct Z Axis MANUAL CONTROL by Software Panel	If this Register = 0, triggering _ManualControlOnAxis_3 on software panel, system will enable Z Axis MANUAL CONTROL ; if this Register ≠ 0, system will not enable Z Axis MANUAL CONTROL when triggering _ManualControlOnAxis_3.	Write/Read
R16108~R16133	Trigger Software Panel _ManualControlOnAxis_4~16 , _ManualControlOffAxis_4~16 Action	Same as above	Read

	Disable Direct 4th~16th Axis MANUAL CONTROL by Software Panel		Write/ Read
R16134	Trigger Software Panel _Reset Action	The register represents _Reset action on software panel. When user triggers _Reset on software panel, this register = 1.	Read
R16135	Disable Direct Reset by Software Panel	If this Register = 0 , triggering _Reset on software panel, system will RESET; if this Register ≠ 0 , system will not RESET when triggering _Reset .	Write/ Read
R16136	Trigger Software Panel _SingleBlockOn, _SingleBlockOff Action	The register represents _SingleBlockOn, _SingleBlockOff action on software panel. When user triggers _SingleBlockOn on software panel, this register = 1. When user triggers _SingleBlockOff on software panel, this register = 0.	Read
R16137	Disable Direct Single Block by Software Panel	If this Register = 0 , triggering _SingleBlockOn on software panel, system will enable Single Block Stop; if this Register ≠ 0 , system will not enable Single Block Stop when triggering _SingleBlockOn .	Write/ Read

R16138	Software Panel Mode Selection	The register represents the argument N of _Mode_N action on software panel, and the action switch modes base on value of N: 1: Edit 2: Auto 3: MDI 4: Jog 5: InJog 6: MPG 7: Home I.e.: Triggering _Mode_2 will set the register =2, and switch to Auto Mode.	Read
R16139	Disable Direct Change Mode Selection by Software Panel	If this Register = 0, triggering _Mode_N on software panel, system will switch Modes; if this Register ≠ 0, system will not switch Modes when triggering _Mode_N.	Write/ Read
R16140	Software Panel Increment Feed	The register represents the argument N of _IncrementFeed_N action on software panel, which decides the magnify ratio of 1st MPG & InJog Example: Triggering _IncrementFeed_10, this register =10 and the magnify ratio become x10. The single movement of InJog increased from 0.001 to 0.10.	Read
R16141	Disable Direct Change Increment Feed by Software Panel	If this Register = 0, triggering _IncrementFeed_N on software panel, system will adjust Magnify Ratio of 1st MPG & InJog; if this Register ≠ 0, system will not adjust Magnify Ratio of 1st MPG & InJog when triggering _IncrementFeed_N.	Write/ Read
R16142	Software Panel Spindle Override	The register represents the argument N of _SpindleOverride_N action on software panel, which decides the Override of Spindle. Example: Triggering _SpindleOverride_10, this register = 50 and the Spindle Override become 50%.	Read

R16143	Disable Direct Change Spindle Override by Software Panel	If this Register = 0, triggering _SpindleOverride_N on software panel, system will adjust spindle override; if this Register ≠ 0, system will not adjust spindle override when triggering _SpindleOverride_N.	Write/Read
R16144	Software Panel Feedrate Override	The register represents the argument N of _FeedrateOverride_N action on software panel, which decides the Override of G01. Example: Triggering _FeedrateOverride_50, this register = 50 and the G01 Override become 50%.	Read
R16145	Disable Direct Change Feedrate Override by Software Panel	If this Register = 0, triggering _FeedrateOverride_N on software panel, system will adjust G01 override; if this Register ≠ 0, system will not adjust G01 override when triggering _FeedrateOverride_N.	W/R
R16146	Software Panel JOG Override	The register represents the argument N of _JOGOverride_N action on software panel, which decides the Override of Jog. Example: Triggering _JOGOverride_50, this register = 50 and the override become 50%.	Read
R16147	Disable Direct Change JOG Override by Software Panel	If this Register = 0, triggering _JOGOverride_N on software panel, system will adjust Jog override; if this Register ≠ 0, system will not adjust Jog override when triggering _JOGOverride_N.	Read / Write
R16148	Software Panel Rapid Traverse Override	The register represents the argument N of _RapidTravelOverride_N action on software panel, which decides the Override of G00. Example: Triggering _RapidTravelOverride_50, this register = 50 and the G00 Override become 50%.	Read

R16149	Disable Direct Change Rapid Traverse Override by Software Panel	If this Register = 0 , triggering _RapidTravelOverride_N on software panel, system will adjust G00 override ; if this Register ≠ 0 , system will NOT adjust G00 override when triggering _RapidTravelOverride_N .	Read / Write
R16150~R16999	Reserved for Software Panel	Reserved for Software Panel Function	
R17000	Reserved for Laser Marking	Reserved for Laser Marking Function	
R17001	SCANNER_OFFSET XOffset	X Offset of laser marking galvanometer module, unit : μm .	Read / Write
R17002	SCANNER_OFFSET YOffset	Y Offset of laser marking galvanometer module, unit : μm .	Read / Write
R17003	Reserved for Laser Marking	Reserved for expansion of Z offset.	
R17004	SCANNER_ANGLE Angle	Galvanometer module scanner mirror rotation angle, unit: 10^{-6} deg. angle > 0 rotate counter-clockwise. angle < 0 rotate clockwise.	Read / Write
R17005	MAX laser frequency(unit: kHz)	Maximum laser frequency (unit: kHz)	Read / Write
R17006	MIN laser frequency(unit: kHz)	Minimum laser frequency (unit: kHz)	Read / Write
R17007	Laser delay time (unit: 0.1 ms)	Laser delay time (unit: 0.1 ms)	Read / Write

R17008	Laser mode selection	Laser mode selection 0: Not selected 1: Auto Mode 2: Marking while data transmitting	Read / Write
R17009	Laser Marking Control Interface	Laser program start running when the Register=1, and the Register is cleared to 0 when program starts. When the corresponding Bit ON, the corresponding function is triggered, and R17009 is automatically cleared to 0. (Do not trigger multiple functions at once) Bit 0: Trigger laser marking machining. Bit 1: Trigger the laser marking reset. (C37 can also trigger a reset)	Read / Write
R17010	Laser end programming request	When the R value is 1, end the machining process of marking-while-data-transmitting, except under synchronous marking mode. After activated, system clear R-value with 0 automatically.	Read / Write
R17011	Enable manual function for galvo meter	Enable manual control of galvo motor R17011= 0: Disabled R17011= 1: Enabled	Read / Write
R17012	Enable manual laser energy	Enable manual laser emitting R17012 = 0: Enabled R17012 = 1: Disabled	Read / Write
R17013	Set manual laser frequency(unit: kHz)	Set manual laser frequency (unit: kHz)	Read / Write

R17014~ R17016	Set manual jog for galvo motor	Setup the manual jog for galvanometer scanner R17014: Axis ID (1:X, 2: Y, 3: Z) R17015: Magnify Ratio 0: 0.1x 1: 1x 2: 10x 3: 100x R17016: Basic Movement, unit: mdeg. Take the deviation value as the current movement.	Read / Write
R17018	Machining cycle mode	0: Single interpretation of the machining file. 1: Repeat the interpretation of the machining file a limited number of times. 2: Repeat the interpretation of the machining file indefinitely until the reset signal (Reset) is triggered. 3: Record mode for the small files (unsupported with 70SB) 4: Play mode for the small files (unsupported with 70SB)	Read / Write
R17019	Machining paths interpretation repeat count	If R17018 is set to 1, the value specified by this R value is used as the counts of repeated interpretations of the processing file. If R17018 is set to 1, and R17019 is set to a non-positive integer, cycle start will trigger the alarm LASERCTRL-006.	Read / Write
R17020	Laser marking override	Setup laser marking feedrate override, override must be > 0 and $\leq 200\%$. R17020 = 100 means feedrate is 100%; R17020 = 150 means feedrate is 150%.	Read / Write
R17021	Part detecting trigger signal	Only applied in conveyer marking mode, an I Bit is used to detect whether the parts on the conveyer passed by, and the I Bit informs the kernel there are parts passing by with R17021. When R17021 turns to 1 from 0 means there are parts being detected. The laser kernel will set R17021 back to 0 automatically after receiving the signal.	Read / Write
R17022	Part detecting trigger signal - macro	Only applied in conveyer marking mode, an I Bit is used to detect whether the parts on the conveyer passed by, and the I Bit informs macro there are parts passing by with R17022. When R17022 turns to 1 from 0 means there are parts being detected. Macro will send the marking data to kernel and set R17022 back to 0 automatically after receiving the signal.	Read / Write

R17023~ R17024	Laser Marking Function Reserved Area	Reserved for Laser Marking Functions	Read / Write
R17025	Laser Marking Synchronous Mode/Tracking Mode Setup	Laser Marking Synchronous Mode/Tracking Mode Setup 0: Disabled 1: Tracking Mode (Product Line)	Read / Write
R17026	Laser Marking Synchronous Mode Setup	Laser Marking Synchronous Mode Setup 0: Disp-Base 1: Time-Base	Read / Write
R17027~ R17030	Laser Marking Function Reserved Area	Reserved for Laser Marking Functions	Read / Write
R17031~ R17040	Assign laser file name	Assign Laser File Name, 10 R values reserved, maximum 39 bytes (Plus 1 ending character) Example: Macro : @117031:= "Laser_Main" Registers: R17031'e's'a'L' R17032 'a'M'_'r' R17033 '\0''n''i'	Read / Write
R17041~ R17042	Laser Source Pulse Width Modulation	R17041: Set to 1, laser source switched to data transmission mode; set to be 0, laser source data transmission mode off. R17042: Pulse width (unit: ns). Range limitation: 0~65535 The 71SA needs to use with the FC-LSMPW to connect to the first piece of the first station.	Read / Write

R17043~ R17045	Reserved for Laser source pulse width setting	Reserved for Laser source pulse width setting	
R17046~ R17099	Reserved for Laser Marking	Reserved for Laser Marking	
R17100	Laser status	Laser status 0 : Not Ready 1 : Ready 2 : Processing 3 : Manual	Read
R17101	Laser mode	Laser mode 0: Not selected 1: Auto Mode 2: Marking while data transmitting	Read
R17103~ R17109	Reserved for Laser Marking Function	Reserved for Laser Marking Function	
R17110	Laser energy command	Laser energy command (0~255)	Read
R17111~ R17113	Laser galvo axis command(unit: mdeg)	Laser galvanometer scanner axis command (unit: mdeg)	Read
R17114~ R17116	Laser galvo axis feedback(unit: mdeg)	Laser galvanometer scanner axis feedback command (unit: mdeg) These Registers display command value if no feedback.	Read
R17117~ R17119	Reserved for Laser Marking	Reserved for Laser Marking	Read

R17120	Conveyor Current Speed	Conveyor Current Speed When the conveyor sensor being set as a normal encoder, it shows the actual feedback; when being set as the virtual speed, it shows the virtual conveyor speed.	Read
R17121~ R17124	Laser Marking Function Reserved Area	Laser Marking Function Reserved Area	Read
R17125	Laser Marking Synchronous Mode/Tracking Mode Setup	Laser Marking Synchronous Mode/Tracking Mode Setup 0: Disabled 1: Tracking Mode (Product Line)	Read
R17126	Laser Marking Synchronous Mode Status	Current Laser Marking Synchronous Mode Status 0: Disp-Base 1: Time-Base	Read
R17127~ R17150	Laser Marking Function Reserved Area	Laser Marking Function Reserved Area	Read
R17151	Laser Energy Display	Decide the laser source type according to parameter 3402 and display the laser energy of each laser source type corresponding R value, unit: %. The corresponding R value of SPI is R880 The corresponding R value of IPG is R53	Read
R17152	Manual Laser On Time Input	The R value is related to the manual laser time of the controller, the laser turns on when the R value is not 0, unit: second.	Read / Write

R17153	Remaining Manual Laser On Time	The R value is related to the remaining manual laser time, the laser turns off when the R value is 0, unit: second.	Read
R17154	Laser Energy Setup	Decide the laser source type according to parameter 3402 and fill the energy into corresponding R values. The corresponding R value of SPI is R880 The corresponding R value of IPG is R53	Read / Write
R17155	Part Rotary Angle Setup	Set the rotary amount of laser marking, the original unit: 10^{-6} deg, change the rotary amount unit to deg. angle > 0 means rotating counterclockwise angle < 0 means rotating clockwise	Read / Write
R17156	Set the grid point number on single side of Grid Distortion Correction Marking (NxN)	The grid point number on single side when calling G533 for Grid Distortion Correction Marking. i.e. When the value is 5, it means to mark a 5x5 grid points. Since the kernel of the Grid Distortion Correction only accepts 3x3, 5x5, 9x9, 17x17, 33x33, total 5 correction options, the value should better set to be 3,5,6,17,33.	Read / Write
R17157	Grid Distortion Correction Marking Range Setup (mm)	The unilateral length (mm) when calling G533 for Grid Calibration Marking.	Read / Write
R17158	Maximum Distortion Amount	The maximum distortion amount of all grid points when doing vision distortion measurements (um).	Read / Write

R17159	Host stopped slave device machining mechanism	When R17159=1, it triggered C37 and run the sub-program O1111. The content of O1111 is controller send HMI_Reset action.	Read / Write
R17160	Set manual laser pulse width (unit: ns)	Manual Laser Pulse Width Setup (Unit: ns)	Read / Write
R17161~R17200	Laser Marking HMI System Reserved Area	Laser Marking HMI System Reserved Area	
R17201~R17210	Woodworking Drilling Machine Occupied R value G54~G57 station cycle start	0: Empty, not processing, or being overwrite to 0 by PLC after reset. 1: The station is ready for machining.	Read / Write
R17211	Woodworking Station Alarm	When system couldn't find the R1720x file, unable to assign, the R value changes to x (x= 1~4).	Read
R17212~R17300	Woodworking Function Reserved Area	Woodworking Function Reserved Area	
R17301~R17350	Five-Axis Function Reserved Area	Five-Axis Function Reserved Area	

R17351~ R17400	Laser Cutting Process Parameters	Laser Cutting Process Parameters, CYC File.	Read / Write
R17401~ R17440	Laser Cutting Servo Correction – Height Voltage Parameters	Laser Cutting Servo Correction – Height Voltage Parameters, CYC File.	Read / Write
R17441~ R17461	Laser Cutting Speed Power Chart	Laser Cutting Speed Power Chart, CYC File.	Read / Write
R17462~ R17500	Laser Cutting Software Panel Parameters (Software Panel)	Laser Cutting Software Panel Parameters (Software Panel)	Read / Write
R17501~ R17548	Laser Cutting Power, Frequency, Servo PLC Calculating Area.	Laser Cutting Power, Frequency, Servo PLC Calculating Area.	Read / Write
R17549~ R17556	Laser Cutting Customized Functions (Tube exchange, Process repeat, Coordinate transformation)	Laser Cutting Customized Functions (Tube exchange, Process repeat, Coordinate transformation)	Read / Write

R17557~ R17600	Laser Cutting Function Reserved Area	Laser Cutting Function Reserved Area	Read / Write
R17601~ R17699	High-Speed Spindle Glass Milling Machine Reserved Area	High-Speed Spindle Glass Milling Machine Reserved Area	
R17700~ R17999	Reserved for CNC	System Reserved Area	
R18000~ R18999	Vision Function Reserved Area	Vision Reserved Area - ref 3.7 System Parameter(R/L) Instructions ²⁰	
R19000~ R19199	Robot Process Package Reserved Area	Robot Process Package Reserved Area	
R19000~ R19199	Robot Process Package Reserved Area	Robot Process Package Reserved Area	
R19200~ R19999	Reserved for CNC	Reserved for system (Consider the robot data coherence when using)	
R20159~ R65535	User define	User-defined	

® marked components mean the signal or Register will be processed instantly.

²⁰ <https://confluence.syntecclub.com.tw/pages/viewpage.action?pageId=94838084>